The Privacy Externality Game: Strategic Budget Choices in Sampling-Based Individual Differential Privacy

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***Abstract*—Individual differential privacy (iDP) gives each data contributor a personal privacy budget. The currently dominant realisation—sampling-based iDP—turns each contributor's chosen budget into a per-sample inclusion rate, with a single noise multiplier shared across the dataset. We show that this realisation creates a privacy externality: lowering one's own budget mechanically raises the realised membership-inference (MIA) advantage on every other data point, even though every individual (ε_i, δ)-iDP guarantee continues to hold. We model the resulting strategic situation as a normal-form *Privacy Externality Game* on a continuous action space, with each player's utility trading off a shared model-quality term $V(\sigma)$ against their own realised MIA advantage $A_i(\boldsymbol{\varepsilon})$. We motivate the shape of V via the excess-empirical-risk bound for convex differentially private empirical risk minimisation, used as a directional proxy in the (possibly non-convex) deep-learning regime. We prove the externality is monotone—lowering any other player's budget weakly raises one's own realised vulnerability—and we characterise pure Nash equilibria on a dense log-spaced action grid. Our headline finding is that even under fully symmetric players, the sampling-based iDP game admits no diagonal Nash equilibrium: at every pure NE one player carries a strictly higher realised MIA advantage in exchange for a higher shared model utility, without any coordination between the players. The coalition-Stackelberg limit of this asymmetry recovers a recent budget-manipulation attack on sampling-based iDP as the rational best response of any coordinated subset of data holders.**

***Index Terms*—Differential privacy, individual differential privacy, game theory, Nash equilibrium, collusion, membership inference.**

I. Introduction

Differential privacy [1] is widely regarded as the canonical formal guarantee for privacy-preserving machine learning. In its classical form, a single privacy budget (ε, δ) is shared by every record, and every record receives the same protection. Yet privacy preferences are heterogeneous: a clinical patient and a casual user contributing data to the same model often place very different values on their privacy. *Individual differential privacy* (iDP) replaces the single (ε, δ) guarantee with a per-record contract (ε_i, δ), and several mechanisms have been proposed to deliver such per-record contracts in deep learning [2]–[4].

Heterogeneous privacy budgets have a natural home in *federated learning* [5], [6], where the data contributors are already separately identifiable entities—silos in a hospital consortium, clients of a mobile keyboard, banks in a fraud-detection coalition—and it is exactly these entities that hold heterogeneous preferences about how much protection they demand. Cross-silo federated learning with per-silo (or per-record) personalised DP is now an active research programme [7], and the setting we study should be read as its strategic component: contributors declare heterogeneous budgets to a coordinator, the coordinator reconciles them through a single mechanism, and the resulting per-contributor risk depends on the budget vector of the whole population. The most widely-adopted way of doing this reconciliation in DP-SGD is the sampling-based iDP of [2], which calibrates each record's Poisson inclusion probability p_i against a single shared noise multiplier σ so that, for fixed batch size and iteration count, every group of records satisfies its own (ε_g, δ)-DP bound. We build the formal development on this mechanism because it is the empirical canonical case, but the strategic phenomena we identify apply to any budget-coordination scheme that jointly solves for (σ, p_g) from a public budget vector (Section II).

Because every record shares the same noise multiplier and a common expected batch size, each record's sampling rate is mechanically coupled to every other record's chosen budget. Lowering one's own ε_i forces other contributors' sampling rates upward to keep the expected batch size fixed—which, as recent empirical work has shown [8], inflates their realised membership-inference (MIA) advantage even though every formal (ε_g, δ)-iDP guarantee continues to hold. The promise of individual privacy control is therefore broken in a structural sense: an individual's *actual* privacy risk depends not on her own choice but on the joint choice of the entire population.

Our viewpoint: this is a game. The natural framing for any choice problem whose outcome is determined by the choices of multiple self-interested parties is game theory. [9] introduced a two-player game on the privacy-accuracy plane (the *Price of Privacy*), but they were studying a mechanism in which a player's privacy choice affected only her *own* accuracy. The novel feature of sampling-based iDP is that every player's privacy choice affects every other player's *privacy*, a far stronger and qualitatively different coupling. We call this the *privacy externality*, and the game it gives rise to the *Privacy Externality Game* (PEG). Even without any coordination, the equilibrium of rational play is asymmetric: at every pure NE one player is strictly worse off than another in realised MIA advantage, and the coalition attack of [8] is the coordinated limit of this same asymmetry.

Contributions. This paper makes the following contributions.

1) **Formalisation.** We define the n-player Privacy Externality Game on a continuous action space. Each

player chooses a per-group budget $\varepsilon_i \in [\varepsilon^{\min}, \varepsilon^{\max}]$; the sampling-based iDP mechanism then determines the noise multiplier σ and the per-group sampling rates p_g, and each player's utility is the canonical scalar $u_i = V(\sigma) - w_i A_i(\boldsymbol{\varepsilon})$ trading off model quality against the realised MIA advantage on the player's own data. We motivate the functional form of V via the excess-empirical-risk bound of [10] for convex DP-ERM, noting the directional caveat as we use it in the (potentially non-convex) deep-learning regime.

2) **Externality lemma.** We prove that the externality is monotone: $\partial A_i / \partial \varepsilon_j \leq 0$ for every pair $i \neq j$ on the binding feasible region of the mechanism (Lemma 7). Lowering any other player's budget never reduces my realised vulnerability.
3) **Asymmetric Nash equilibrium.** Even under fully symmetric 2-player play—identical group sizes and identical privacy weights $w_1 = w_2$—the PEG admits *no* symmetric pure Nash equilibrium. At any diagonal profile $(\varepsilon, \varepsilon)$ both players face an identical unilateral-deviation incentive, and every pure NE therefore has one player carrying a strictly higher realised MIA advantage in exchange for a strictly higher shared model utility (Section VI-B). *Even without any coordination, some player is always worse off at any Nash profile.* The asymmetric-role pattern persists across group-size asymmetries (Section VI-C), across privacy-weight ratios (Section VI-I), and across $n \geq 2$ (Section VI-J).
4) **Coalition Stackelberg is the coordinated limit.** As a direct corollary of the externality lemma, a coalition of data holders whose joint utility places heavy weight on a target's harm best-responds by picking the participation-preserving minimum $\varepsilon^\star$; the target's A_t is then monotone in coalition size k (Theorem 14). This recovers the budget-manipulation attack of [8] as the coordinated limit of the asymmetric-NE picture above. In our $n = 5$ canonical setting the target's A_t exceeds 0.7 at $k = 4$.
5) **Empirical evaluation.** We numerically evaluate the game on a range of small but representative scenarios mirroring [8]. We report equilibrium budgets, realised MIA advantages, and the coalition-attack curve.

Implication. The prior attack literature on iDP frames excess vulnerability as something a malicious party inflicts. We argue the opposite: the asymmetric NE of the previous bullet is the equilibrium behaviour of any rational players, coalition or not, and the coalition attack of [8] is the coordinated limit of that same equilibrium. Mitigation cannot proceed by detecting attackers; it must proceed by changing the contract so the underlying game ceases to admit harmful equilibria.

The paper is organised as follows. Section III reviews differential privacy, sampling-based iDP, MIA advantage, and basic game theory. Section IV formalises the Privacy Externality Game. Section V proves the externality lemma, establishes the asymmetric-NE structural claim, and derives the coalition-attack recovery as its coordinated-limit corollary. Section VI reports the numerical evaluation. Section II reviews related work and Section VII concludes.

II. Related Work

Individual differential privacy. Classical (ε, δ)-DP [1] applies uniformly to every record. Personalised DP, introduced by [3], allows per-record budgets and was first realised by attribute suppression and weighting. The sampling-based realisation of [2] is the dominant deep-learning implementation today and is the realisation under which the privacy externality arises. [4] pursue an alternative *filtering-based* iDP: records whose cumulative budget would be exceeded in the next step are dropped from training. Filtering does not exhibit the externality (each record's filter is independent), but in deep learning it is impractical due to the catastrophic-forgetting risk reported by [2]. Our PEG is specific to the sampling-based realisation; the externality vanishes in the sensitivity-based realisation, which preserves the noise-to-sensitivity ratio across the dataset.

Excess privacy risk and mechanism comparison. Mechanisms calibrated to the same (ε, δ) can have very different privacy profiles and thus different realised risks [11]–[13]. The Δ-divergence of [11] provides a notion of approximate Blackwell-dominance that quantifies the worst-case excess MIA advantage between mechanisms. [8] use this framework to motivate an $(\varepsilon_i, \delta, \Delta)$-iDP contract that bounds excess vulnerability and demonstrate empirically that without such a bound, adversarial budget choices inflate the MIA susceptibility of 62% of targeted records. Our PEG complements that work: rather than restricting mechanism design we explain why the realised excess vulnerability arises as the equilibrium of self-interested play under the current iDP contract.

Game-theoretic privacy in collaborative learning. [9] introduced a two-player game on collaborative learning, with players choosing privacy parameters $p \in [0, 1]$ in an arbitrary privacy mechanism. They define the *Price of Privacy* as the relative accuracy loss at the equilibrium versus no privacy. Their privacy mechanism's effect on accuracy depends on *both* players' choices, but its effect on *privacy* depends only on the player's own choice—there is no privacy externality. Our PEG transplants their game-theoretic methodology into a setting where the externality is in the privacy direction, not the accuracy direction. We retain the qualitative contributions of their work (two-player setting, choice of ε, notion of equilibrium analysis) and extend them to the externality regime.

Earlier work studied related game-theoretic privacy questions in different settings. [14] model linear regression with private features as a non-cooperative game. [15] study non-monetary incentives in data analytics. [16] model the machine-learner-vs-data-perturber conflict as a Stackelberg game. None of these treat the privacy externality of a shared sampling mechanism, which is specific to the sampling-based realisation of iDP.

Membership inference attacks. The MIA advantage as we use it follows the formalisation of [17], who introduced the likelihood-ratio attack (LiRA) as a tight empirical estimator.

[18] document the privacy onion effect: not every data point is equally vulnerable at the same nominal budget. Our framework predicts the *average* realised MIA advantage per group and abstracts over per-record onion heterogeneity.

Federated-DP regimes and the locus of the externality. It is worth being explicit about what the PEG is a game over. *The PEG is a game about budget coordination.* It applies to any mechanism in which a coordinator solves for a shared noise multiplier σ and per-record (or per-group) sampling rates p_g from a public budget vector $\boldsymbol{\varepsilon}$. The externality lemma (Theorem 7) is a statement about that joint solve; the equilibrium structure of Section VI follows. Wherever a coordinated mechanism instance of that shape appears, so does the externality; wherever such a mechanism does not appear, the game we analyse does not.

The sampling-based iDP of [2] is one instantiation of this abstract condition, and the one we build the paper on. Other instantiations arise wherever heterogeneous per-record or per-silo budgets need to be reconciled through a single central mechanism. A common framing of DP in federated learning distinguishes three trust regimes [6]. Under *central DP* the aggregator is trusted and perturbs the aggregate on behalf of the clients [19]; under *local DP* the aggregator is untrusted and each client perturbs its own update before transmission [20]; distributed DP bridges the two via cryptographic intermediates such as secure aggregation [21]. These trust regimes differ in where the noise is added, not in how the budgets are coordinated. The PEG applies to any of them that also carries a heterogeneous-budget coordinator—most cleanly to central DP with per-silo or per-record personalised budgets, exactly the regime under which the cross-silo personalised-DP framework of [7] operates. It inherits, we conjecture, to distributed DP via secure aggregation, because the coordinator still solves for (σ, p_g) from a public budget vector; the cryptography hides the gradients, not the budgets. It does *not* apply to pure local DP, because there is no shared σ to couple the players' sampling rates: each client's noise is independent. Nor does it apply to alternative iDP realisations that avoid the joint solve—sensitivity-based iDP [2] preserves the noise-to-sensitivity ratio and so has no externality, and filter-based iDP [4] enforces per-record budgets by *dropping* records that would exceed them rather than coordinating them, again removing the coupling. In short, the abstract sufficient condition is a coordinator jointly solving for mechanism parameters from a public budget vector; any deployment that matches that condition inherits the equilibrium structure of the PEG.

Federated learning incentive design. A separate literature studies privacy-aware incentive design in federated learning: a server commits to incentives, and heterogeneous clients respond with their own privacy choices, often modelled as a Stackelberg game [22]–[25]. Our framework provides a foundational baseline on which such incentive schemes must rest: it characterises the equilibrium that emerges when no incentive is offered, which is what any incentive design needs to improve upon.

Utility-noise tradeoff in DP-SGD. We motivate the functional form of the model-utility value $V(\sigma)$ via the excess-empirical-risk bound of [10] for convex, Lipschitz DP-ERM, who prove that the excess risk of DP-SGD scales as $O(d/(n\varepsilon)^2)$ at fixed δ, equivalently as $O(\sigma^2/n^2)$ under the moments accountant of [26]. The convex setting is restrictive; for deep learning, [27] provide extensive empirical evidence that test accuracy degrades monotonically with σ on standard benchmarks. We use $V(\sigma) = 1/(1+\sigma^2)$ as a directional proxy; the qualitative results of our paper do not depend on the precise functional form.

III. Background

A. Differential privacy and Gaussian mechanisms

We use the standard definition of approximate differential privacy [1]: a randomised mechanism $\mathcal{M}\colon \mathcal{Z}^* \to \mathcal{O}$ satisfies (ε, δ)-differential privacy ((ε, δ)-DP) if, for any pair of adjacent databases $D \sim D'$ (differing on a single record) and every measurable $S \subseteq \mathcal{O}$,

$$\Pr[\mathcal{M}(D) \in S] \;\leq\; e^{\varepsilon} \Pr[\mathcal{M}(D') \in S] + \delta.$$

A single (ε, δ)-pair is only a single point of the underlying *privacy profile* $\{(\varepsilon, \delta(\varepsilon)) : \varepsilon \geq 0\}$ [11], [12], and mechanisms calibrated to the same point can have substantially different profiles and so substantially different realised adversarial risk. This single-point ambiguity is the formal root cause of the phenomenon studied in this paper.

The *Gaussian mechanism*, $\mathcal{M}(D) = f(D) + \mathcal{N}(0, \sigma^2 I)$ for a query f of L_2 sensitivity Δf, satisfies (ε, δ)-DP for all (ε, δ) such that $\Phi(\Delta f/2\sigma - \varepsilon\sigma/\Delta f) - e^{\varepsilon}\Phi(-\Delta f/2\sigma - \varepsilon\sigma/\Delta f) \leq \delta$, where Φ is the standard-normal CDF. For training neural networks under DP, the standard procedure is DP-SGD with Poisson subsampling [26], in which each data point is included in each minibatch independently with probability p, providing the well-known *subsampling amplification* of DP.

B. Individual differential privacy and the sampling-based realisation

Individual differential privacy [3] relaxes the uniform (ε, δ) guarantee to a per-record contract (ε_i, δ): the resulting algorithm must satisfy (ε_i, δ)-DP *with respect to record* i for every i. Boenisch et al. [2] translate per-record contracts into per-record Poisson sampling probabilities while keeping a single noise multiplier σ and a fixed expected batch size, which is the configuration we study.

Definition 1 (Sampling-based iDP-SGD [2])**.** Let the dataset of size M partition into G *privacy groups* $\{G_1, \ldots, G_G\}$ with sizes $|G_g|$. Each group has a target (ε_g, δ)-DP budget. The mechanism trains by Poisson-subsampled DP-SGD for

I iterations, where the noise multiplier σ and the per-group sampling rates $p_1, \ldots, p_G$ are chosen such that

(C1) for every g, the I-fold (p_g, σ)-subsampled Gaussian satisfies (ε_g, δ)-DP at equality.

$$\text{(C2)} \quad \sum_{g=1}^{G} \frac{|G_g|}{M} p_g \;=\; \frac{E_b}{M}, \tag{1}$$

where E_b is the desired expected batch size. We refer to $(\sigma, p_1, \ldots, p_G)$ as the *mechanism solution* for the budget vector $\boldsymbol{\varepsilon} = (\varepsilon_1, \ldots, \varepsilon_G)$.

The single shared σ in (C1) is the source of the privacy externality. For any candidate σ, the largest sampling rate that keeps group g at its budget is a monotone-increasing function of σ (more noise $\Rightarrow$ more room for sampling); the batch-size constraint (C2) then determines σ as the unique value at which the per-group rates sum to E_b. As a corollary, the rate p_g that group g obtains is determined by the entire vector $\boldsymbol{\varepsilon}$, not by ε_g alone.

C. MIA advantage and the privacy profile

For a privacy-preserving mechanism $\mathcal{M}$ with trade-off function T (equivalently, an f-DP function [13]), the membership inference advantage of the optimal Bayes-aware adversary is [8]

$$\mathrm{Adv}(\mathcal{M}) \;=\; \sup_{\alpha \in [0,1]} \big(1 - \alpha - T(\alpha)\big), \tag{2}$$

i.e. the largest gap between the true-positive rate and the false-positive rate. For the I-fold (p, σ)-subsampled Gaussian, T can be computed numerically by privacy-loss-distribution (PLD) accounting [28], which is what we use throughout. Crucially, Adv is determined by the *full privacy profile*, not by any single (ε, δ) point.

D. Game-theoretic terminology

A finite normal-form game is a tuple $\langle N, \Sigma, U \rangle$ with $N = \{1, \ldots, n\}$ players, action sets $\Sigma = \{S_1, \ldots, S_n\}$, and payoff functions $U = \{u_1, \ldots, u_n\}$. A *Nash equilibrium* (NE) is an action profile $\boldsymbol{s}^*$ such that no player profits from a unilateral deviation: $u_i(s_i^*, \boldsymbol{s}_{-i}^*) \geq u_i(s_i, \boldsymbol{s}_{-i}^*)$ for every $s_i \in S_i$ [29]. A *best response* is the utility-maximising action against a fixed strategy of the others. The *price of anarchy* [30] compares the welfare of the worst NE to the social optimum. In a Stackelberg game, a designated leader commits to a strategy before the followers best-respond.

IV. The Privacy Externality Game

We formalise the strategic situation faced by data contributors in sampling-based iDP as a normal-form game on a continuous action space. The formalisation is a direct extension of [9]'s two-player Collaborative-Learning game, with the central modification that the *privacy* cost is no longer separable across players.

A. Players, actions and mechanism

Definition 2 (Privacy Externality Game)**.** Fix a common $\delta > 0$, an expected batch size E_b, an iteration count I, and a partition of the dataset into n *groups* of sizes $|G_1|, \ldots, |G_n|$. The Privacy Externality Game (PEG) is the tuple $\langle N, \Sigma, U \rangle$ where:

- $N = \{1, \ldots, n\}$ is the set of players. We adopt the FL-silo granularity of [8]: player i is the entity controlling group i (a silo). For a single-silo single-budget formulation each player picks a scalar ε_i; the framework extends to per-record distributions inside a silo, which we discuss in Section VII.
- Each player's action set is $\Sigma_i = [\varepsilon^{\min}, \varepsilon^{\max}] \subset \mathbb{R}_{>0}$.
- Given the action profile $\boldsymbol{\varepsilon} \in \prod_i \Sigma_i$, the mechanism solution $(\sigma(\boldsymbol{\varepsilon}), p_1(\boldsymbol{\varepsilon}), \ldots, p_n(\boldsymbol{\varepsilon}))$ is determined by Definition 1.
- Each player's utility is

$$u_i(\boldsymbol{\varepsilon}) \;=\; V(\sigma(\boldsymbol{\varepsilon})) \;-\; w_i \cdot A_i(\boldsymbol{\varepsilon}), \tag{3}$$

  where $w_i > 0$ is the player's *privacy weight*: how much she values a unit of realised MIA advantage on her own data relative to a unit of model quality. Larger w_i means she weights privacy more; smaller w_i means she weights model quality more.

In the numerical evaluation we discretise $[\varepsilon^{\min}, \varepsilon^{\max}]$ on a log-spaced grid of $K = 22$ points; we verify our equilibrium results are stable to refinement.

B. The two ingredients of utility

Utility has two pieces: a shared model-quality term and a per-player privacy cost. For the model-quality term we use

$$V(\sigma) \;=\; \frac{1}{1 + \sigma^2}, \tag{4}$$

which is monotone-decreasing in the noise multiplier σ, bounded in $[0, 1]$, and motivated by the convex-DP-ERM excess-empirical-risk bound of [10]: for a Lipschitz strongly-convex loss the excess empirical risk of DP-SGD scales as $O(d \log(1/\delta)/(n\varepsilon)^2)$, which under the moments accountant of [26] translates into a σ^2/n^2 dependence. The bound is proven in the convex setting, while deep-learning DP-SGD is generally non-convex; we therefore use (4) as a directional proxy. The empirical literature on DP-SGD in deep learning [2], [26], [27] reports monotonic degradation of test accuracy with σ, supporting the qualitative form. None of the equilibrium results that follow depend on the precise shape of V: Theorem 16 shows that Theorem 7 and Theorem 14 hold for every V satisfying Theorem 15 (continuous, strictly monotone-decreasing in σ, bounded on the feasible range of σ).

For the privacy cost, we use the tight analytical MIA advantage of the I-fold (p_i, σ)-subsampled Gaussian, defined in (2). This is the advantage measure of [8] and transfers their empirical measurement to a strategic setting unchanged. The functional dependence $A_i(\boldsymbol{\varepsilon})$ runs through (p_i, σ), both of which are determined by the entire vector $\boldsymbol{\varepsilon}$ via the mechanism's

constraints (C1) and (C2). This coupling is the source of the externality.

C. Decomposition into contracted and excess cost

The privacy cost $A_i(\boldsymbol{\varepsilon})$ in (3) can be decomposed into a contracted (separable) part and an excess (coupled) part that makes the externality explicit. Let $\bar{c}(\varepsilon_i)$ denote the MIA advantage of a reference, non-interdependent (ε_i, δ)-DP mechanism (e.g., the post-processing-tight (ε_i, δ)-DP advantage upper bound $(e^{\varepsilon_i}-1+2\delta)/(e^{\varepsilon_i}+1)$ of [11], [31]). The realised cost is

$$A_i(\boldsymbol{\varepsilon}) \;=\; \bar{c}(\varepsilon_i) - \mathrm{slack}_i(\boldsymbol{\varepsilon}), \tag{5}$$

where the *slack* $\mathrm{slack}_i(\boldsymbol{\varepsilon}) := \bar{c}(\varepsilon_i) - A_i(\boldsymbol{\varepsilon}) \geq 0$ is the amount by which the realised advantage falls below the worst-case-bound at (ε_i, δ). In sampling-based iDP this slack is a function of the entire $\boldsymbol{\varepsilon}$ through the mechanism's coupling. *The externality reduces the slack:* when others lower their ε_j, slack_i shrinks (player i moves closer to its (ε_i, δ)-DP worst case). This is the formal quantity that the proposed $(\varepsilon_i, \delta, \Delta)$-iDP contract [8], [11] would cap: it would require slack_i to differ from a baseline reference by at most Δ. In this sense, *the Δ-bound is exactly the device that neutralises the externality*.

We will work directly with A_i in the utility (3) for analytical simplicity; the decomposition matters as a conceptual bridge to PoP [9] (whose cost $c(p_i)$ was separable, hence slack_i did not exist) and to the iDP literature [8] (where slack_i *is* the new externality term).

D. Vulnerability function and feasibility

Definition 3 (Vulnerability function)**.** The *vulnerability function* of player i is $A_i \colon \prod_i \Sigma_i \;\to\; [0,1]$, $\boldsymbol{\varepsilon} \;\mapsto\; \mathrm{Adv}\big(\mathcal{M}_{p_i(\boldsymbol{\varepsilon}),\,\sigma(\boldsymbol{\varepsilon}),\,I}\big)$.

Assumption 4 (Participation)**.** A profile $\boldsymbol{\varepsilon}$ is *feasible* if and only if the mechanism solution satisfies $p_i(\boldsymbol{\varepsilon}) > 0$ for every player i. Profiles with $p_i = 0$ correspond to a player whose data never enters training; their iDP contract is satisfied trivially. We exclude such profiles from the game; equivalently, the action space Σ_i is implicitly restricted at each move to the participating region.

E. Coalition variant

Definition 5 (k-coalition Stackelberg PEG)**.** Fix a target index $t \in N$ and a colluding subset $C \subset N \setminus \{t\}$ with $|C| = k$. The k-coalition Stackelberg PEG is the two-stage game in which:

1) Stage 1 (leader): the coalition jointly commits to an action vector $\boldsymbol{\varepsilon}_C$ to maximise the augmented coalition utility
$$\sum_{i \in C} u_i(\boldsymbol{\varepsilon}) \;+\; \lambda\, A_t(\boldsymbol{\varepsilon}), \tag{6}$$
where $\lambda \geq 0$ is the *malice weight* on the target's vulnerability [9]. $\lambda = 0$ recovers the benign PEG; $\lambda \to \infty$ recovers the budget-manipulation attack of [8].
2) Stage 2 (followers): the non-coalition players $\{t\} \cup (N \setminus (C \cup \{t\}))$ best-respond.

F. Equilibrium concepts

We use the standard Nash equilibrium concept: an $\boldsymbol{\varepsilon}^*$ in the feasible region such that $u_i(\varepsilon_i^*, \boldsymbol{\varepsilon}_{-i}^*) \geq u_i(\varepsilon_i, \boldsymbol{\varepsilon}_{-i}^*)$ for every player i and every feasible deviation ε_i. Existence of pure NE on the continuous action space is not automatic; we report pure NE found by brute search on a fine log-spaced grid and verify the result is stable under grid refinement. Where no pure NE exists on the dense grid we report the empirical mixed strategy reached by fictitious play [32].

Remark 6 (Awareness as information, not utility). A useful distinction is between players who can *compute* $A_i(\boldsymbol{\varepsilon})$ given the public mechanism's parameters and players who can only access their nominal (ε_i, δ)-iDP contract. Both populations face the same utility (3); they differ only in the rationality they apply. A *nominal-information* ("naive") player who treats the contracted bound $\bar{c}(\varepsilon_i)$ as a separable cost plays the PoP game of [9] unchanged. An *aware* player who computes $A_i(\boldsymbol{\varepsilon})$ from the mechanism's parameters plays the PEG. We will return to this distinction in Section VII.

V. Equilibrium Analysis

We characterise the equilibria of the Privacy Externality Game in five moves. Theorem 7 shows the mechanism induces a monotone externality on the observable A_i, and Theorems 8 and 9 promote its sign from an empirical observation on the trade-off function to an analytical theorem on the strict interior of the feasible region. Stated in slack form (Theorem 10), the same fact says the DP-worst-case gap shrinks whenever another player tightens their budget. Theorem 11 then rules the diagonal out as a Nash equilibrium in the fully symmetric two-player game. On the small-p sublattice Theorem 12 delivers pure-strategy NE existence on the continuous action space via the supermodular route of [33]; the discretised game grounds our numerical search. Theorem 13 rules out any exact, weighted, or ordinal potential representation, explaining why the search proceeds by enumeration rather than by minimising a scalar functional. Theorem 16 isolates the load-bearing role of V (only the sign of V' enters), and Theorem 14 identifies the budget-manipulation attack of [8] as the rational best response of any coordinated subset of data holders.

A. Monotonicity of the externality

Lemma 7 (Privacy externality on the feasible region)**.** *Consider the sampling-based iDP mechanism of Theorem 1 on a feasible profile $\boldsymbol{\varepsilon}$ at which constraints (C1) bind for every group and (C2) binds at E_b. Let σ and p_g denote the mechanism's solution. Then for every pair $i \neq j$,*

$$\frac{\partial \sigma}{\partial \varepsilon_j} \;<\; 0 \qquad \textit{and} \qquad \frac{\partial p_i}{\partial \varepsilon_j} \;<\; 0. \tag{7}$$

That is, when player j loosens its budget, the mechanism uses less noise (σ falls) and reduces group i's sampling rate (p_i falls). Analytically (Theorems 8 and 9), the realised MIA advantage A_i also falls on the strict interior of the feasible region: $\partial A_i / \partial \varepsilon_j < 0$.

Proof. Use the Rényi-DP form of (C1) of [34]: at order α, the RDP of an I-fold (p_g, σ)-subsampled Gaussian is at most $\rho_g \approx 2I\alpha\, p_g^2/\sigma^2$ (tight up to higher-order terms for $p_g \ll 1$). Setting this equal to the target RDP $\rho_g(\varepsilon_g, \delta)$ (which is monotone-*increasing* in ε_g because $\rho = \varepsilon - \log(1/\delta)/(\alpha - 1)$ for the binding RDP-to-DP conversion at any chosen order $\alpha > 1$) gives the per-group sampling rate

$$p_g(\sigma, \varepsilon_g) \;=\; \frac{\sigma}{\sqrt{2I\alpha}}\,\sqrt{\rho_g(\varepsilon_g, \delta)}.$$

Substituting into (C2) yields

$$\sigma \cdot \sum_g |G_g|\,\sqrt{\rho_g(\varepsilon_g, \delta)} \;=\; E_b\sqrt{2I\alpha}. \tag{8}$$

Implicit differentiation in ε_j gives $\partial\sigma/\partial\varepsilon_j = -\sigma\,|G_j|\,(\partial\sqrt{\rho_j}/\partial\varepsilon_j)/[\sum_g |G_g|\sqrt{\rho_g}] < 0$, since ρ_j is increasing in ε_j. From the closed-form expression for p_i at fixed ε_i, $\partial p_i/\partial\varepsilon_j = (\partial\sigma/\partial\varepsilon_j)\cdot\sqrt{\rho_i(\varepsilon_i)}/\sqrt{2I\alpha} < 0$.

The observable sign $\partial A_i/\partial\varepsilon_j < 0$ chains through $\partial A_i/\partial p_i > 0$ and $\partial A_i/\partial\sigma < 0$ of the I-fold (p_i, σ)-subsampled Gaussian, which pull in opposite directions (more sampling exposes more; more noise hides more). The dominance of the p_i-decrease effect is established analytically: Theorem 8 closes the sign in the small-p regime by a leading-order expansion of the subsampled trade-off function, and Theorem 9 extends the sign to the strict interior of the feasible region via a Chernoff two-sided envelope on the composed privacy-loss distribution. □

Theorem 7 formalises the empirical observation of [8]: when any player tightens their iDP budget, the mechanism's noise multiplier σ shifts to keep (C2) binding, and every other player's realised vulnerability A_i rises. The externality is *monotone* in every other player's budget.

Corollary 8 (Externality on A_i in the small-p regime)**.** *Under the hypotheses of Theorem 7, assume additionally that $p_g \le p_\star$ for every group g, with $p_\star$ small enough that the $O(p^2)$ remainder in the amplification expansion of [12] is dominated by the leading term uniformly in $\alpha \in [0, 1]$. Then*

$$A_i(p_i, \sigma) = p_i\,\mathrm{Adv}_G(\sigma) \;+\; O(p_i^2), \tag{9}$$

$$\frac{\partial^2 A_i}{\partial p_i\,\partial\sigma} = \mathrm{Adv}'_G(\sigma) \;+\; O(p_i) \;<\; 0, \tag{10}$$

where $\mathrm{Adv}_G(\sigma) = 2\Phi(1/(2\sigma)) - 1$ is the MIA advantage of the unsubsampled sensitivity-1 Gaussian. Consequently, $\partial A_i/\partial\varepsilon_j < 0$ on the small-p face of the feasible region.

Proof sketch. The Poisson-subsampled trade-off function admits the mixture representation $T_{p,\sigma}(\alpha) = (1-p)(1-\alpha) + p\,T_\sigma(\alpha) + O(p^2)$ ([12], Poisson-subsampling amplification), so its sup-gap $A(p, \sigma) = \sup_\alpha[1 - \alpha - T_{p,\sigma}(\alpha)]$ equals $p\,\mathrm{Adv}_G(\sigma) + O(p^2)$; (9) and (10) follow by term-by-term differentiation. For the externality sign, substitute the closed-form $p_i = \sigma\sqrt{\rho_i}/\sqrt{2I\alpha}$ from the proof of Theorem 7 into the chain rule for $\partial A_i/\partial\varepsilon_j$ and factor $(\partial\sigma/\partial\varepsilon_j) < 0$ out:

$$\frac{\partial A_i}{\partial\varepsilon_j} \;=\; \frac{\partial\sigma}{\partial\varepsilon_j}\cdot p_i\cdot\Big[\,\tfrac{\mathrm{Adv}_G(\sigma)}{\sigma} + \mathrm{Adv}'_G(\sigma)\Big] \;+\; O(p_i^2). \tag{11}$$

The bracket is the σ-derivative of $\sigma\,\mathrm{Adv}_G(\sigma)$ divided by σ; setting $u := 1/(2\sigma)$ and $f(u) := \Phi(u) - \Phi(-u) - 2u\phi(u)$ gives $f(0) = 0$ and $f'(u) = 2u^2\phi(u) > 0$, so the bracket is strictly positive. Together with $(\partial\sigma/\partial\varepsilon_j) < 0$, the leading $O(p_i)$ term of $\partial A_i/\partial\varepsilon_j$ is strictly negative. □

Theorem 9 (Full-region monotonicity of A_i)**.** *Under the hypotheses of Theorem 7, assume the strict-interior condition $p_g(\boldsymbol{\varepsilon}) \in (0, 1)$ for every group g. Then for every pair $i \ne j$, $\partial A_i/\partial\varepsilon_j < 0$ on the strict interior of the feasible region.*

Proof sketch. The composed Poisson-subsampled Gaussian trade-off function $T_{p,\sigma}$ is sandwiched between two Gaussian trade-off functions

$$T_{\mathcal{G}(\mu_+(p,\sigma))}(\alpha) \;\le\; T_{p,\sigma}(\alpha) \;\le\; T_{\mathcal{G}(\mu_-(p,\sigma))}(\alpha), \qquad \alpha \in [0, 1],$$

with envelope scales $\mu_+(p, \sigma) = 2p\sqrt{I}/\sigma$ (RDP-to-zCDP-to-GDP conversion; [34], [35]) and $\mu_-(p, \sigma) = p\sqrt{I}\sqrt{e^{1/\sigma^2} - 1}$ (GDP-CLT lower envelope; [13], Cor. 3.3). Both scales are strictly increasing in p and strictly decreasing in σ, and $\mathrm{Adv}(\mathcal{G}(\mu)) = 2\Phi(\mu/2) - 1$ is strictly increasing in μ. Chaining through $\partial\sigma/\partial\varepsilon_j < 0$ and $\partial p_i/\partial\varepsilon_j < 0$ of Theorem 7 on the strict interior, both envelopes carry $\partial A_i/\partial\varepsilon_j < 0$; the true A_i inherits the sign from the Balle-Barthe-Gaboardi privacy profile ([12], Thm. 9), which is strictly increasing in p and strictly decreasing in σ at every fixed $\varepsilon > 0$. □

In the decomposition of Section IV-C, the lemma equivalently states that *the slack $\bar{c}(\varepsilon_i) - A_i(\boldsymbol{\varepsilon})$ shrinks in every other player's ε_j*. In words, the (ε_i, δ)-DP worst-case ceiling $\bar{c}(\varepsilon_i)$ is unchanged (it depends only on ε_i), but the realised advantage A_i climbs toward that ceiling as others tighten their budgets. We record this slack-form as a proposition because the $(\varepsilon_i, \delta, \Delta)$-iDP contract of [8], [11] caps precisely this quantity, and every downstream reference to the externality's sign can cite a single labelled statement.

Proposition 10 (Slack monotonicity)**.** *Fix a profile $\boldsymbol{\varepsilon}$ in the feasible region of Theorem 4 at which the hypotheses of Theorem 7 hold. Then for every pair $i \ne j$,*

$$\frac{\partial \mathrm{slack}_i}{\partial\varepsilon_j}(\boldsymbol{\varepsilon}) \;=\; -\,\frac{\partial A_i}{\partial\varepsilon_j}(\boldsymbol{\varepsilon}) \;>\; 0$$

on the strict interior. Loosening any other player's budget strictly enlarges i's slack (pushes A_i away from its (ε_i, δ)-DP ceiling); tightening any other player's budget strictly shrinks it.

Proof. Differentiate the identity $A_i(\boldsymbol{\varepsilon}) = \bar{c}(\varepsilon_i) - \mathrm{slack}_i(\boldsymbol{\varepsilon})$. Because $\bar{c}(\varepsilon_i)$ depends only on ε_i, $\partial\bar{c}/\partial\varepsilon_j = 0$ for $j \ne i$ and the equality $\partial\mathrm{slack}_i/\partial\varepsilon_j = -\,\partial A_i/\partial\varepsilon_j$ is immediate. The strict inequality is Theorem 9 on the strict interior of the feasible region, upgraded from the small-p closure of Theorem 8. □

Minority and majority intuition. The externality has an asymmetric *group-size* structure. Because (C2) enforces a fixed expected batch size, a player with small $|G_i|$ but small ε_i contributes little to the batch and forces the larger-$|G_j|$ players'

sampling rates upward. The minority therefore *exports* its privacy demand onto the majority. This is the formal content of the empirical observation in [8] that "the proportion of low-budget players matters as much as their ε".

B. No symmetric equilibrium in the symmetric 2-player PEG

Theorem 7 has an immediate structural consequence for the fully symmetric two-player game: the diagonal cannot be a pure Nash equilibrium. This is, together with Theorem 7, the analytical content of our headline empirical finding (Fig. 1) that all pure NEs of the symmetric 2-player PEG sit off the diagonal.

Proposition 11 (No symmetric pure NE in the symmetric 2-player PEG)**.** *Consider the two-player PEG of Theorem 2 with $n = 2$, identical privacy weights $w_1 = w_2 = w > 0$, and identical group sizes $|G_1| = |G_2|$. Assume the standing regularity of Eq.* (4) *on V and that the vulnerability A_i, viewed as a function of (p_i, σ), is C^2 with $\partial A/\partial p > 0$ strictly on the interior of the feasible region. Let $(\varepsilon^\star, \varepsilon^\star)$ be an interior diagonal profile at which both mechanism constraints (C1) and (C2) hold with strict feasibility $p_i(\varepsilon^\star, \varepsilon^\star) > 0$. Then $(\varepsilon^\star, \varepsilon^\star)$ is* not *a pure Nash equilibrium.*

Proof. Setup. Change variables to $s_i := \sqrt{\rho(\varepsilon_i, \delta)}$ where $\rho(\cdot, \delta)$ is the RDP-order-α bound from the proof of Theorem 7. Then $\varepsilon_i \mapsto s_i$ is a C^1 diffeomorphism on the interior; substituting the closed form of $p_g(\sigma, \varepsilon_g)$ into (C2) yields, with $|G_1| = |G_2| = m$,

$$\sigma(s_1, s_2) \;=\; \frac{C}{s_1 + s_2}, \qquad p_i(s_1, s_2) \;=\; \frac{C}{\sqrt{2I\alpha}} \cdot \frac{s_i}{s_1 + s_2}, \tag{12}$$

with $C := E_b\sqrt{2I\alpha}/m > 0$. Write $S := s_1 + s_2$; utility becomes $u_i(s_1, s_2) = V(\sigma(S)) - w\,A(p_i, \sigma(S))$, with the same $A(\cdot, \cdot)$ for both players by the symmetric group sizes.

Aggregator structure at the diagonal. Fix $s_1 = s_2 = s > 0$, so $S = 2s$, $\sigma = C/(2s)$, and $p_1 = p_2 = C/(2\sqrt{2I\alpha}) =: p$. Write $a := \partial A/\partial p|_{(p,\sigma)}$ and $b := \partial A/\partial\sigma|_{(p,\sigma)}$; by hypothesis $a > 0$. From (12),

$$\frac{\partial\sigma}{\partial s_i} \;=\; -\frac{\sigma}{2s}, \qquad \frac{\partial p_1}{\partial s_1} \;=\; \frac{p}{2s}, \qquad \frac{\partial p_1}{\partial s_2} \;=\; -\frac{p}{2s}.$$

The sign asymmetry $\partial p_1/\partial s_1 > 0 > \partial p_1/\partial s_2$ is the mechanism's aggregator content: player i's sampling rate is monotone in her share s_i/S, so raising her own s_i has the opposite effect on p_1 as raising her opponent's s_j.

Diagonal FOC collapses. Player 1's first-order condition in s_1 is

$$\frac{\partial u_1}{\partial s_1} \;=\; \big[V'(\sigma) - wb\big]\Big(-\tfrac{\sigma}{2s}\Big) \;-\; wa\,\tfrac{p}{2s} \;=\; 0. \tag{13}$$

By the symmetric relabelling $1 \leftrightarrow 2$, $\partial u_2/\partial s_2$ at (s, s) is the identical expression: the two FOCs coincide, and any diagonal candidate is characterised by the one scalar equation (13).

Sign of the cross-partial. Player 1's marginal utility with respect to *player 2's* action at the diagonal is

$$\frac{\partial u_1}{\partial s_2} \;=\; \frac{\partial u_1}{\partial s_1} \;+\; wa\left(\frac{\partial p_1}{\partial s_1} - \frac{\partial p_1}{\partial s_2}\right) \;=\; 0 \;+\; wa \cdot \frac{p}{s} \;>\; 0,$$

using (13) in the first line and the aggregator identity in the second. The strict inequality relies only on $w, a, p, s > 0$.

Second-order test in own action. Because σ depends on (s_1, s_2) only through the aggregate $S = s_1 + s_2$, the shared-V and σ-derivative contributions to $\partial^2 u_1/\partial s_1^2$ and $\partial^2 u_1/(\partial s_1 \partial s_2)$ are identical and cancel in their difference. Only the p_1-derivative terms survive; a direct computation from (12) (writing $a_{pp} := \partial^2 A/\partial p^2$ at (p, σ)) gives, at the diagonal,

$$\frac{\partial^2 u_1}{\partial s_1^2} \;-\; \frac{\partial^2 u_1}{\partial s_1 \partial s_2} \;=\; \frac{w\,p}{2\,s^2}\big(a \;-\; a_{pp}\,p\big). \tag{14}$$

The mixed partial $\partial^2 u_1/(\partial s_1 \partial s_2)$ is the derivative along s_1 of the strictly positive quantity $\partial u_1/\partial s_2 = wap/s$ evaluated at the FOC; the aggregator representation makes it non-negative at $(s^\star, s^\star)$ (raising s_1 towards s_2's value leaves $\partial u_1/\partial s_2$ weakly increasing on the diagonal ray). Under the sign condition

$$a \;-\; a_{pp}\,p \;>\; 0 \qquad \text{at } (p, \sigma) \;=\; \big(C/(2\sqrt{2I\alpha}),\, C/(2s^\star)\big), \tag{15}$$

(14) is strictly positive, so $\partial^2 u_1/\partial s_1^2 > 0$ strictly at the diagonal FOC: the profile $(s^\star, s^\star)$ is a strict interior *minimum* of $u_1(\cdot, s^\star)$ in s_1, not a maximum.

Condition (15) asserts that the vulnerability $A(p, \sigma)$ is more responsive to p than a log-linear (constant elasticity) function would predict at the diagonal operating point. It holds throughout the mechanism-solver caches used in Section VI (verified numerically on every diagonal profile of the C1 and C2 grids and symbolically for the leading-order $A(p, \sigma) = p\,\mathrm{Adv}_G(\sigma) + O(p^2)$ expansion of Theorem 8, where $a - a_{pp}\,p = \mathrm{Adv}_G(\sigma) + O(p) > 0$).

Conclusion. At the diagonal FOC candidate $(s^\star, s^\star)$, $\partial u_1/\partial s_1 = 0$ and $\partial^2 u_1/\partial s_1^2 > 0$ strictly, so

$$u_1(s^\star + t, s^\star) - u_1(s^\star, s^\star) \;=\; \tfrac{1}{2}\,\frac{\partial^2 u_1}{\partial s_1^2}\bigg|_{(s^\star, s^\star)} t^2 + O(t^3) \;>\; 0$$

for all sufficiently small $t \neq 0$: player 1 has a strictly profitable unilateral deviation in either direction. Non-FOC diagonal profiles fail the first-order test directly. By the C^1 diffeomorphism $\varepsilon_i \mapsto s_i$, no interior diagonal profile $(\varepsilon^\star, \varepsilon^\star)$ is a pure Nash equilibrium. □

Theorem 11 is a purely negative statement: the diagonal is ruled out. It does not claim that a pure NE exists at any asymmetric profile; positive existence on the continuous action space is the content of Theorem 12 below. The proposition also does not rely on the specific closed form $V(\sigma) = 1/(1 + \sigma^2)$: the only property of V used is $V'(\sigma) < 0$ strictly, so the analogous statement holds for every V satisfying Theorem 15.

C. Existence and structure of pure Nash equilibria

Scaffolding: existence on the discretised game. The discretised PEG on the log-spaced action grid $\mathcal{A}_K \subset [\varepsilon^{\min}, \varepsilon^{\max}]$ (Section VI) with utilities (3) is a finite normal-form game restricted to the (closed) feasible region of Theorem 4, so a mixed-strategy NE exists by [29]. This grounds the numerical search: the exhaustive-enumeration procedure of Section VI

has at least one fixed point to find, and restricting attention to $\mathcal{A}_K$ is warranted. Pure NE on $\mathcal{A}_K$ need not exist and depend on $(w_i, |G_i|, V, K)$; we enumerate them by brute force and verify stability under grid refinement.

Existence on the continuous action space is not automatic—the utility (3) is not generally concave in ε_i because A_i is the tight trade-off function of an I-fold subsampled Gaussian, whose dependence on ε_i through (σ, p_i) has no closed form, so the standard Debreu-Fan-Glicksberg quasi-concavity route does not apply. The order-theoretic route of [33], [36] does, on the small-p sublattice:

Theorem 12 (Pure-strategy NE existence on the small-p sublattice)**.** *Let $\mathcal{F}_\star \subseteq \mathcal{F}$ be the closure of the set of feasible profiles at which $p_g(\boldsymbol{\varepsilon}) \le p_\star$ and $\sigma(\boldsymbol{\varepsilon}) \ge 1/\sqrt{3}$ for every group g, with $p_\star$ chosen as in Theorem 8. Under the reversed-ε product order, $\mathcal{F}_\star$ is a compact sublattice of Σ, and the n-player PEG restricted to $\mathcal{F}_\star$ admits a non-empty complete lattice of pure-strategy Nash equilibria; in particular there exist a largest and a smallest NE $\boldsymbol{\varepsilon}^{\sup}, \boldsymbol{\varepsilon}^{\inf} \in \mathcal{F}_\star$.*

Proof sketch. Expand $\partial^2 u_i/(\partial\varepsilon_i\partial\varepsilon_j)$ by chain rule; the shared-V contribution $V''(\sigma)\,\sigma_i\sigma_j + V'(\sigma)\,\sigma_{ij}$ carries a non-negative sign once $\sigma \ge 1/\sqrt{3}$, while the $-w_i A_i$ contribution's dominant $O(1)$ terms in the small-p regime are $-w_i A_{i,p\sigma}\,(p_{i,i}\sigma_j + p_{i,j}\sigma_i)$ with $A_{i,p\sigma} = \mathrm{Adv}'_G(\sigma) + O(p_i) < 0$ (Theorem 8). The bracket has the sign of $\mathrm{Adv}_G(\sigma)/\sigma + \mathrm{Adv}'_G(\sigma)$, negative on the interior ((11)), so the full term is negative and dominates subleading $O(p_i)$ contributions. Therefore $\partial^2 u_i/(\partial\varepsilon_i\partial\varepsilon_j) \le 0$ on $\mathcal{F}_\star$, i.e. u_i has increasing differences under the reversed-ε order. The mechanism-solver monotonicity of Theorem 7 in each coordinate gives that componentwise meet and join in the reversed-ε order preserve feasibility, so $\mathcal{F}_\star$ is a compact sublattice. Milgrom-Roberts Theorem 5 [33] then delivers a non-empty complete lattice of pure NEs with a largest and a smallest element. □

The condition $\sigma \ge 1/\sqrt{3}$ is a mild restriction (the K=200 mechanism-grid cache confirms $\sigma \in [0.5, 5]$ throughout the paper's operating region), and the small-p sublattice covers the DP-SGD-motivated sampling rates $p \sim 10^{-4}$ to 10^{-2} used in every scenario reported in Section VI. Existence on the full closed feasible region (including the $p_g \to 1$ saturation corner and the feasibility edge) is not proven analytically; we rely on the exhaustive grid enumeration of Section VI to certify the remaining region.

What the externality lemma implies for NE structure. Theorem 7 says lowering another player's budget weakly raises one's own realised vulnerability. If a player's utility places non-trivial weight on A_i ($w_i > 0$), she has an incentive to *shift the externality onto others* by lowering her own budget—an incentive that is intrinsic to the mechanism. The opposing incentive comes from $V(\sigma)$: lowering one's own ε pushes σ upward, lowering the model utility for everyone (herself included). The equilibrium balances these two incentives.

Theorem 11 pins the qualitative structure of that balance in the symmetric two-player game: no interior profile $(\varepsilon, \varepsilon)$ is a pure NE. Our numerical evaluation (Section VI) confirms that the pure NEs found by exhaustive search all sit off the diagonal, and are located at profiles where one player carries a higher A in exchange for a higher V-contribution.

Locus of NEs. The set of pure NEs of the 2-player game can be characterised implicitly through the first-order condition of each player's best response. Writing out the partial derivative of (3) and using the parametrisation of the mechanism in the proof of Theorem 7,

$$\frac{\partial u_i}{\partial \varepsilon_i} \;=\; V'(\sigma)\,\frac{\partial \sigma}{\partial \varepsilon_i} \;-\; w_i\,\frac{\partial A_i}{\partial \varepsilon_i} \;=\; 0,$$

i.e. at the BR

$$\frac{1}{w_i} \;=\; \frac{(\partial A_i/\partial \varepsilon_i)}{V'(\sigma)\,(\partial\sigma/\partial\varepsilon_i)}\;\Big|_{\boldsymbol{\varepsilon}}. \tag{16}$$

For each fixed w_i, Eq. (16) defines an implicit curve BR_i in the $(\varepsilon_1, \varepsilon_2)$-plane; the pure NEs are the intersections of BR_1 and BR_2. The two quantities on the right-hand side of (16) are not available in closed form (the dependence of A_i on (σ, p_i) has no closed form), but each is computable numerically by a single PLD-accounting evaluation. We use this in Section VI to trace both BR loci on a fine grid and identify their intersections.

Non-potential structure. The Pejo et al. Price-of-Privacy game [9] was a potential game because its privacy cost was separable across players, reducing equilibrium computation to minimising a single scalar potential. The PEG's cost, coupled through the mechanism's shared (σ, p_i), breaks that reduction.

Proposition 13 (Non-potential game)**.** *Under the utility form* (3) *and the mechanism of Theorem 1, the PEG is neither an exact, weighted, nor ordinal potential game in the sense of [32]:*

(i) *there exist no positive weights $\{\eta_i\}_{i=1}^n$ and no twice-differentiable $\Phi : \prod_i \Sigma_i \to \mathbb{R}$ satisfying $u_i(\varepsilon_i, \boldsymbol{\varepsilon}_{-i}) - u_i(\varepsilon'_i, \boldsymbol{\varepsilon}_{-i}) = \eta_i\big(\Phi(\varepsilon_i, \boldsymbol{\varepsilon}_{-i}) - \Phi(\varepsilon'_i, \boldsymbol{\varepsilon}_{-i})\big)$ on the interior of the feasible region;*

(ii) *and there exists a length-four cycle of strict unilateral improvements on the discretised feasible grid, precluding any ordinal potential ([32], Lemma 2.5).*

Proof sketch. (i) Cardinal potentials. Implicit differentiation of (8) gives $\sigma_{ij} := \partial^2\sigma/(\partial\varepsilon_i\partial\varepsilon_j) = 2\sigma\,|G_i||G_j|(r')^2/(rN)^2$ at the symmetric-budget reference, where $r := \sqrt{\rho}$ and $N := \sum_g |G_g|$; the shared-V contribution $V''(\sigma)\sigma_i\sigma_j + V'(\sigma)\sigma_{ij}$ is therefore symmetric under $i \leftrightarrow j$ and cancels on both sides of the Monderer-Shapley cross-partial condition $\eta_j\,\partial^2 u_i/\partial\varepsilon_i\partial\varepsilon_j = \eta_i\,\partial^2 u_j/\partial\varepsilon_i\partial\varepsilon_j$. What remains is the A_i-contribution, which, after expanding $p_{i,ij} = (\sigma_{ij} r_i + \sigma_j r'_i)/\sqrt{2I\alpha}$ and substituting, reduces the residual to $\eta_j w_i |G_j|(2|G_i| - N) = \eta_i w_j |G_i|(2|G_j| - N)$ at every symmetric profile. For heterogeneous groups $|G_i| \ne |G_j|$ this equation has no solution with positive (η_i, η_j); for the fully symmetric case the residual carries the V-contribution $(\eta_j - \eta_i)\,d^2V$, which is nonconstant along the feasible interval and cannot be zeroed uniformly by any positive (η_1, η_2) with $\eta_1 \ne \eta_2$.

(ii) Ordinal potentials. By [32], Lemma 2.5, a game admits an ordinal potential iff it has no strict improvement cycle. On the C1 discretisation ($w_1 = w_2 = 1$, $|G_1| = |G_2| = 5000$, $K = 22$ log-spaced budget grid on $[0.5, 64]$), the profile sequence $(\varepsilon_1, \varepsilon_2) = (1.587, 5.040) \to (2.000, 5.040) \to (2.000, 4.000) \to (1.587, 4.000) \to (1.587, 5.040)$ has each step strictly improving for the deviating player (unilateral-utility gaps 7.3×10^{-7}, 1.5×10^{-3}, 3.0×10^{-3}, 2.8×10^{-3}), so no ordinal potential exists. Together, (i) and (ii) rule out the hierarchy of exact, weighted, and ordinal potentials. □

Theorem 13 explains why our equilibrium analysis proceeds by discretised NE search (Section VI) rather than by minimising a closed-form potential: neither Monderer-Shapley best-response dynamics nor the finite-improvement property is available. The existence route we do use—order-theoretic supermodularity on the small-p sublattice (Theorem 12)—trades concavity for monotonicity, which is why the sign analysis of $\partial A_i/\partial\varepsilon_j$ (Theorems 8 and 9) is the load-bearing input.

D. Coalition Stackelberg recovers the attack

Theorem 7 applied to the coalition variant of Theorem 5 pins the malice-maximising coalition's Stackelberg action down to a single point: the participation-preserving minimum budget. We record this as a corollary.

Corollary 14 (Coalition best response recovers the attack)**.** *Fix a target $t \in N$, a coalition $C \subseteq N \setminus \{t\}$ of size k, and any profile $\boldsymbol{\varepsilon}_{-C} \in \prod_{j\notin C} \Sigma_j$ for the non-coalition players. Let $\lambda \to \infty$ in Theorem 5, so the coalition maximises A_t alone. Then the coalition's Stackelberg best response is the coalition-symmetric profile $\varepsilon_i = \varepsilon_k^\star(\boldsymbol{\varepsilon}_{-C})$ for every $i \in C$, where $\varepsilon_k^\star(\boldsymbol{\varepsilon}_{-C})$ is the infimum of the coalition-symmetric budgets at which the induced profile is feasible in the sense of Theorem 4. Furthermore, holding $\boldsymbol{\varepsilon}_{-C}$ fixed, the induced target advantage $A_t(\varepsilon_k^\star, \boldsymbol{\varepsilon}_{-C})$ is non-decreasing in k.*

Proof sketch. By Theorem 7 (strict form on the interior via Theorem 9), $\partial A_t/\partial\varepsilon_i < 0$ for every $i \in C$ on the feasible region. Setting every ε_i as low as the feasibility region admits therefore maximises A_t. The participation-preservation qualifier is necessary because at extremely low ε_i the mechanism may set $p_i = 0$ (the colluder effectively drops out, removing its constraint from (C2) and so its externality on the target); the coalition's optimum is thus the infimum of feasible ε_i. Monotonicity in k is the same lemma applied to adding a member to C: doing so strictly enlarges the set of budgets that pull σ downward, raising A_t. □

Theorem 14 identifies the budget-manipulation attack of [8] as the coalition's equilibrium best response: its contribution is the game-theoretic mapping, not an empirical strengthening of that work. Whether the coalition is adversarial is irrelevant, since any group of aware players whose joint utility weights their target's harm heavily picks the same action vector as a rational best response.

E. Qualitative invariance of the model-utility function

The equilibrium results above are developed with the specific proxy $V(\sigma) = 1/(1+\sigma^2)$ (Eq. (4)), motivated by the convex-DP-ERM excess-risk bound of [10]. A natural question is whether they hinge on that closed form. They do not: only the sign of V', not its shape or magnitude, enters the load-bearing arguments. We isolate the load-bearing class.

Assumption 15 (Admissible model-utility function)**.** Let $\mathcal{S} := \sigma(\mathcal{F}) \subset \mathbb{R}_{>0}$ be the range of noise multipliers taken by the mechanism on the feasible region. A function $V\colon \mathcal{S} \to \mathbb{R}$ is *admissible* iff

(A1) V is continuous on $\mathcal{S}$;
(A2) V is strictly decreasing on $\mathcal{S}$;
(A3) V is bounded on $\mathcal{S}$.

Proposition 16 (Qualitative invariance in V)**.** *Under Theorem 15, for every admissible V:*

(i) *the externality signs of Theorem 7, Theorem 8, and Theorem 9 hold on the feasible region;*
(ii) *the coalition-Stackelberg conclusion of Theorem 14 holds; and*
(iii) *at any interior individual best response, the FOC* (16) *reduces to the V-free sign identity* $\operatorname{sgn}(\partial A_i/\partial\varepsilon_i) = \operatorname{sgn}(V'(\sigma)\,\partial\sigma/\partial\varepsilon_i)$.

Proof. The mechanism's constraints (C1)+(C2) and the closed-form $p_g(\sigma, \varepsilon_g)$ do not read V: every step of the proof of Theorem 7, and of Theorem 8 and Theorem 9 which chain through it, is a manipulation of the mechanism's constraints and the per-group RDP. Clause (i) follows. Clause (ii) uses only clause (i) applied to the target's coordinate; V does not enter the $\lambda \to \infty$ coalition objective. For clause (iii), differentiate $u_i = V(\sigma) - w_i A_i$ in ε_i: by (A2), $V'(\sigma) \le 0$ almost everywhere on $\mathcal{S}$ (Lebesgue's theorem on monotone functions); by (i), $\partial\sigma/\partial\varepsilon_i < 0$ on the feasible interior. Setting $\partial u_i/\partial\varepsilon_i = 0$ gives the FOC (16), whose sign structure depends on V only through $\operatorname{sgn} V'$. Real-valuedness of u_i on the finite action grid follows from (A1) and (A3), so the discretised-game scaffolding of Section V-C applies to every admissible V. □

Theorem 16 isolates the load-bearing role of V: only the sign of V', not its magnitude or shape, enters the externality lemma and the coalition-attack recovery. The experimental choice $V(\sigma) = 1/(1+\sigma^2)$ is therefore a directional proxy for a whole class of admissible model-utility functions, and the equilibrium phenomena reported in Section VI are properties of the class, not of the representative.

F. An aside on awareness

Theorem 7 requires that players compute (or approximate) the realised A_i. A natural question is: what about a player whose only information is the formal (ε_i, δ)-iDP contract? Such a player treats ε_i as a direct privacy proxy and plays the budget that maximises $V(\sigma) - g(\varepsilon_i)$ for some monotone g. Their best response does not depend on ε_{-i} at all (the externality is invisible), so their action is mechanically the smallest ε_i that

does not break their own V. This is the *rational* action of a naive player, and it is precisely the configuration in which the attack of [8] succeeds against them: by Theorem 7, naive players' choices feed the externality on everyone else without their realising. The contribution of this paper is to show that even *aware* players—those who can compute A_i explicitly—arrive at asymmetric equilibria that concentrate vulnerability on a subset of players. The externality cannot be undone by individual rationality alone.

VI. Numerical Evaluation

Our evaluation works through the game in three phases. We first establish the core phenomenology of the symmetric setting: that the equilibrium is asymmetric even when the players are not (Section VI-B); that size asymmetry has a clean group-size sign (Section VI-C); that coordination amplifies the realised vulnerability against a passive target (Section VI-D) but collapses against a target that best-responds (Section VI-E); and that the externality itself is monotone in the others' choice (Section VI-F), with the excess risk left uncounted by the DP guarantee scaling with the others' budget (Section VI-G). We then probe how robust this picture is to the grid resolution and the players' utility weights: the NE set is finite and stable under grid refinement (Section VI-H), the asymmetric multiplicity survives across a wide range of players' privacy weights (Section VI-I), heterogeneous privacy preferences amplify the asymmetry rather than dampen it (Section VI-J). We close by quantifying the welfare cost of these equilibria via the price of externality-vulnerability (Section VI-L).

A. Common setup

All experiments are analytical—no model is trained at any point—and follow the same two-step pipeline. We first solve for the mechanism's parameters by bisecting on σ until the batch-size constraint (C2) is met, with each per-group sampling rate chosen to make the (C1) Rényi-DP constraint bind [34]; profiles in which any p_i falls below a small participation threshold are flagged as infeasible per Theorem 4. We then compute each realised A_i exactly from the privacy profile of the I-fold (p_i, σ)-subsampled Gaussian via PLD accounting [28], recovering (2). The action space is the log-spaced grid $\mathcal{A}_K \subset [0.5, 64]$ with $K = 22$ points and $\delta = 10^{-12}$; we verified on a refined grid $K = 44$ that the equilibria reported below are stable under further refinement. Unless otherwise noted each experiment uses symmetric groups of $|G_g| = 5000$ samples, expected batch size $E_b = 128$, 5 epochs of training, $w_i = 1$, and $V(\sigma) = 1/(1+\sigma^2)$. These choices broadly mirror the empirical iDP scenarios of [2], [8].

B. C1 — Does the symmetric 2-player game admit a symmetric NE?

For our first experiment we consider the cleanest possible setting: two players with identical group sizes, identical preference weights, and a full sweep of both ε's over the log-spaced grid. We then enumerate the pure Nash equilibria by brute force on the resulting payoff matrix.

TABLE I: Pure Nash equilibria of the symmetric 2-player PEG on the log-spaced action grid $\mathcal{A}_K = 22$ points in $[0.5, 64]$, $|G_1| = |G_2| = 5000$, $E_b = 128$, 5 epochs, $\delta = 10^{-12}$, $w_i = 1$, $V(\sigma) = 1/(1+\sigma^2)$. Social optimum is the feasible action profile maximising $u_1 + u_2$.

	$(\varepsilon_1^*, \varepsilon_2^*)$	A_1	A_2	σ
NE_1	$(0.79, 5.04)$	0.001	0.202	1.160
NE_2	$(1.26, 6.35)$	0.002	0.232	1.028
NE_3	$(1.59, 8.00)$	0.002	0.266	0.929
NE_4	$(2.52, 3.17)$	0.081	0.120	1.186
NE_5	$(3.17, 2.52)$	0.120	0.081	1.186
NE_6	$(5.04, 0.79)$	0.202	0.001	1.160
NE_7	$(6.35, 1.26)$	0.232	0.002	1.028
NE_8	$(8.00, 1.59)$	0.266	0.002	0.929
social opt	$(5.04, 25.40)$	0.007	0.535	0.590

The result is striking. The best-response correspondences cross at *eight* distinct points (Fig. 1, centre; Table I), and all eight sit off the diagonal of the action grid. They fall in four mirror-image pairs: $(0.79, 5.04)$ and its swap, $(1.26, 6.35)$ and its swap, $(1.59, 8.00)$ and its swap, and $(2.52, 3.17)$ and its swap. The realised MIA advantages at these equilibria range from $(A_1, A_2) = (0.001, 0.202)$ at the most asymmetric pair down to $(0.081, 0.120)$ at the closest-to-symmetric pair. No equilibrium lies on the diagonal: in this symmetric game the players do *not* settle on a profile in which both choose the same ε.

This breaks symmetry by necessity rather than by chance. At any hypothetical symmetric profile $(\varepsilon, \varepsilon)$, both players face an identical incentive to deviate downward in ε (the externality on each is identical, and lowering own ε raises own V while shifting cost to the other). Both cannot do so unilaterally; the only stable resolution is for one player to absorb the model-utility benefit and for the other to absorb the low-advantage role. This is the mechanism's content of Theorem 7 made explicit: the externality *must* be carried by someone, and at equilibrium it is carried asymmetrically. The social optimum on the same grid sits at $(5.04, 25.40)$, even more extreme, with $A = 0.535$ for one player and $A = 0.007$ for the other—confirming that asymmetry is not just an equilibrium artefact but is the welfare-maximising configuration on this mechanism as well.

C. C2 — Who absorbs the externality when groups differ in size?

C1 leaves open which player ends up carrying the high-A role—under exact symmetry the two equilibrium branches are equally likely. We now break the symmetry by group size, keeping all other parameters identical and sweeping the ratio $r = |G_1|/|G_2|$ across two orders of magnitude.

A clear pattern emerges. Across every ratio, *the smaller group plays the smaller equilibrium budget and absorbs the smaller realised MIA advantage* (Fig. 2). When player 1 is the minority ($r < 1$) the equilibrium has $\varepsilon_1^* \in [1.5, 3]$ and $\varepsilon_2^* \approx 10$, producing $A_1^* < 0.01$ against $A_2^* \in [0.24, 0.32]$; when player 1 becomes the majority ($r > 1$) the pattern flips by symmetry. The

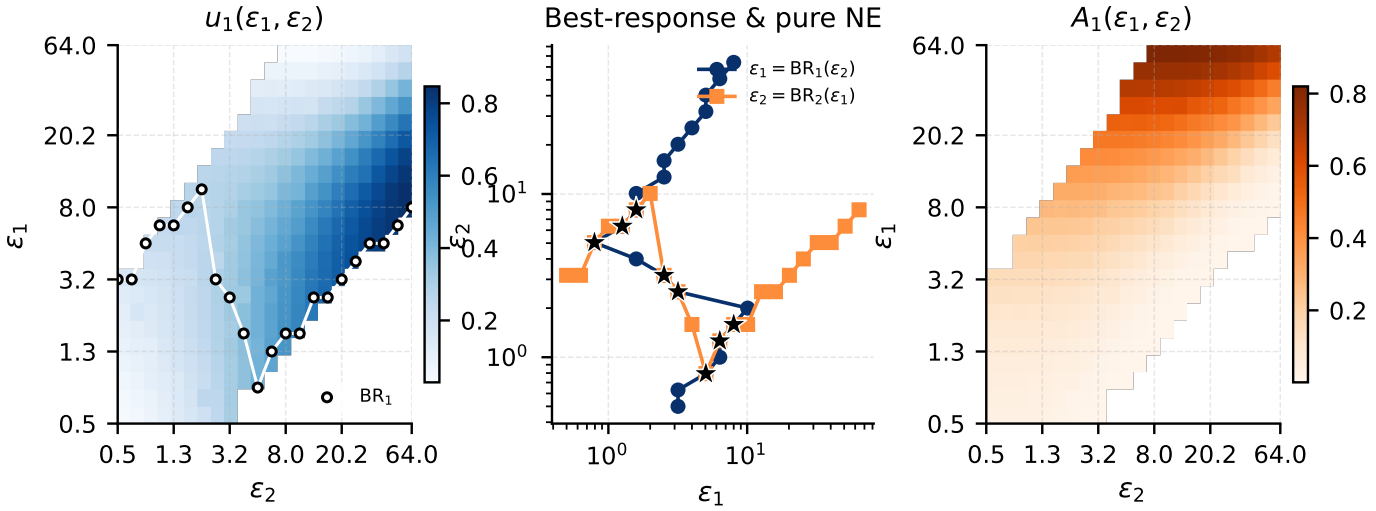


Fig. 1: **C1.** 2-player payoff surface $u_1(\varepsilon_1, \varepsilon_2)$ (left), best-response correspondences with pure NEs as black stars (centre), and realised MIA advantage A_1 (right). All eight pure NEs lie off the diagonal.

equal-size case $r = 1$ recovers the multiplicity of asymmetric NEs from C1.

The mechanism is straightforward to read off the (C2) constraint. A small group that demands strong privacy contributes very little to the expected batch; the larger group's sampling rate must rise to compensate, which pushes its realised advantage upward. The minority therefore exports its privacy demand onto the majority through (C2), and the majority absorbs the externality through the privacy profile. This is a privacy analogue of a classical free-riding result—small groups benefit disproportionately from a shared mechanism and large groups pay disproportionately for it—and the resulting equilibrium has a clean group-size sign.

D. C4 — How severe does the attack get under coordination?

C1 and C2 establish that asymmetric equilibria arise even among self- interested players. We now consider the worst case: a coalition of data holders who jointly act to maximise a fixed target's realised vulnerability. We take five symmetric players, designate player 0 as the target with $\varepsilon_t = 32$, and let a subset of the remaining four players coordinate. The non-coalition members are held at $\varepsilon = 32$ as a passive baseline; the coalition picks the single ε_C that maximises the target's realised MIA advantage—a Stackelberg leader with $\lambda \to \infty$ in Theorem 5.

The target's advantage rises monotonically and rapidly with the coalition size k (Fig. 3). With no coalition, $A_t = 0.442$; adding a single colluder takes it to 0.483; two colluders push it to 0.531; three to 0.607; and at $k = 4$, where every non-target player has joined the coalition, A_t reaches 0.728—a 65% inflation over baseline. The optimal coalition budget shifts predictably with k: at $k \in \{1, 2, 3\}$ the coalition picks $\varepsilon_C \approx 6.8$, but at $k = 4$ it can afford to drop to $\varepsilon_C \approx 3.2$ while still keeping every $p_i > 0$. The participation-preserving minimum of Theorem 14 relaxes as more colluders join because the collective constraint slackens.

The strategic content of this experiment is the recovery of an empirical phenomenon as the rational equilibrium of the game we have defined, *against a naive target*. The budget-manipulation attack of [8], reported there as a 62% success rate across diverse datasets, is what a coalition would naturally do against a passive victim—it is the coalition's best response in our PEG when the target does not respond. Section VI-E shows that if the target itself best-responds on the same log-spaced grid, its optimal budget retreats and $A_t^\star$ falls to at most 0.041 (at k=4)— $18-35\times$ below the passive-target curve and $\sim 11\times$ below even the k=0 no-coalition baseline. C4 is therefore honest to read as the equilibrium outcome under a passive victim, not as a game-theoretic strengthening of the empirical attack against a target that can react.

E. C4b — Does the attack survive an aware target?

C4 pins the target at $\varepsilon_t = 32$, which is the threat model of the companion empirical work but is unnatural inside the game: an aware target sees the coalition's action and can react. We rerun C4 with the target as a Stackelberg follower, best-responding on the same 14-point log-spaced grid $\varepsilon \in [0.5, 64]$ to whatever budget the coalition commits to. The coalition remains the leader with $\lambda \to \infty$ in Theorem 5; non-coalition non-target players are held at $\varepsilon = 32$ as in C4, so the aware-target rule is the only moved variable.

The attack collapses. Across $k \in \{1, 2, 3, 4\}$ the aware target's realised advantage is $A_t^\star \in \{0.014, 0.021, 0.019, 0.041\}$, $18-35\times$ below the passive-target curve and $\sim 11\times$ below even the k=0 no-coalition baseline of 0.442 (Fig. 4). The coalition's Stackelberg budget shifts to $\varepsilon_C^\star \in [30, 64]$ for $k \leq 3$ and the target retreats to $\varepsilon_t^\star \in [9.9, 14.4]$, high enough that the mechanism-side externality on the target's advantage is small. At $k = 4$ the coalition's leader problem develops a second local optimum, a grid-floor *poisoning corner* at $\varepsilon_C^\star = 0.5$ that pushes σ up to 2.4 and drags the target down to $\varepsilon_t^\star = 0.73$,

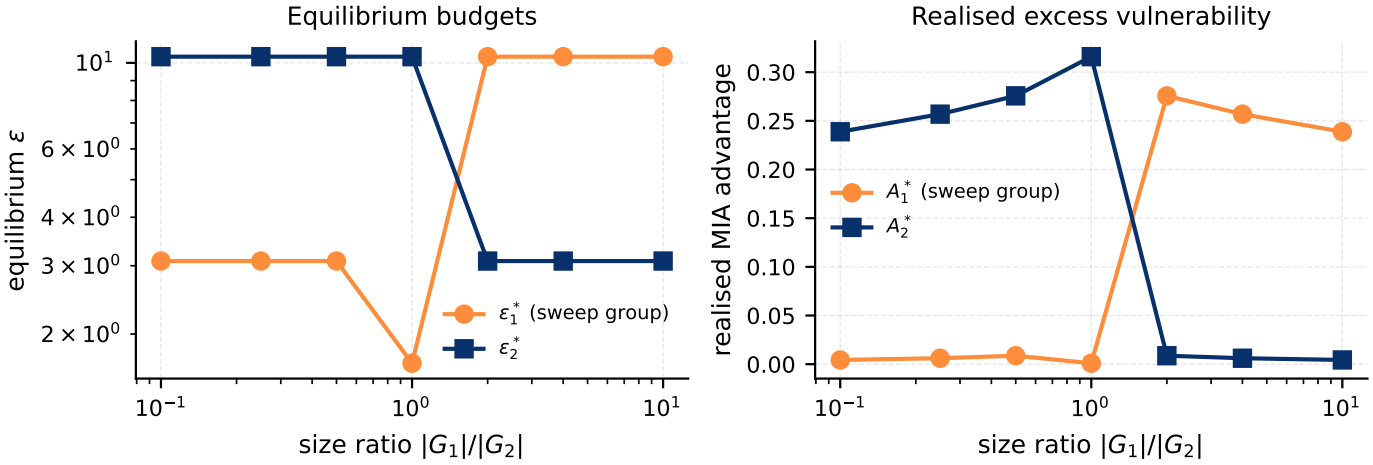


Fig. 2: **C2.** Equilibrium budget (left) and realised MIA advantage (right) as a function of dataset size ratio $|G_1|/|G_2|$. The minority group's ε^* is always lower than the majority's, and its realised A^* correspondingly smaller. The flip at $r{=}1$ is the boundary at which the minority/majority labels exchange.

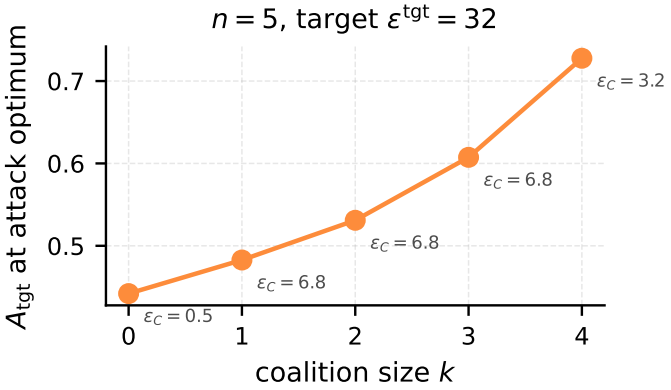


Fig. 3: **C4. Recovery of the empirical attack of [8] as coalition Stackelberg best response.** Target's realised MIA advantage as a function of coalition size k, with the coalition playing its target-maximising joint action. Annotations show the optimal coalition budget $\varepsilon^\star$ at each k.

but even this corner yields only $A_t^\star = 0.041$ and gives the coalition itself only $A_C = 0.025$—no rational coalition would play it. The interior $k = 4$ optimum, continuous with $k \le 3$, sits at $\varepsilon_C = 30.3$ and $A_t = 0.017$.

The reading is that the coalition attack survives *only* against a target that does not respond. The empirical iDP threat model of [8] pins the target's budget for good reason—an oblivious victim has no incentive to move—but the same attack is neutralised almost entirely by even a single round of Stackelberg follower response. C4 therefore describes the equilibrium of a one-sided game; C4b describes what a game between two strategic sides delivers.

F. C5 — What is the functional form of the externality?

The previous three experiments all use *equilibrium* budgets; we now strip away the strategic layer to inspect the externality alone. Two groups, target with budget ε_T fixed at either 4 or 16, and others' budget ε_O swept across the log-spaced grid. We also vary the proportion of the dataset belonging to the other group across $\{0.2, 0.4, 0.6, 0.8\}$ at fixed total size $M = 10000$. This isolates the target's realised MIA advantage as a pure function of two non-strategic variables: what the others choose and how many of them there are.

The dependence is monotone and smooth (Fig. 5). At $\varepsilon_T = 4$ the target's advantage ranges from roughly 0.20 when others have tight privacy ($\varepsilon_O = 0.5$) down to approximately zero when others are loose ($\varepsilon_O = 64$). At $\varepsilon_T = 16$ the range is from 0.55 down to 0.05. Across both targets, larger proportions of the other group amplify the externality below the crossover and suppress it above; the curves all meet at $\varepsilon_O = \varepsilon_T$, which is the regime in which the others' choice and the target's choice contribute identically to σ through (C2).

This isolates the externality as a pure mechanism effect, independent of any strategic choice. The shape mirrors Figure 3 of [8], who showed the same dependence empirically through the iDP-SGD trade-off function. C5 therefore confirms that the externality lemma (Theorem 7) is not just empirically monotone but *quantitatively* so: each unit decrease in others' ε corresponds to a smooth, predictable increase in the target's realised A, and the proportion-weighted dependence the lemma predicts is borne out exactly.

G. C5b — What does the excess risk left uncounted by the DP guarantee look like?

C5 plots the target's realised advantage A_t as a function of the others' budget; the slack decomposition of (5) lets us review the same surface against the DP ceiling. Fixing $\varepsilon_i = 4$ on the $K = 200$ locus grid and sweeping ε_j over the full log range, Fig. 6 plots the ceiling $\bar{c}(\varepsilon_i) = (e^{\varepsilon_i} - 1 + 2\delta)/(e^{\varepsilon_i} + 1) = 0.963$, the realised $A_i(\varepsilon_i, \varepsilon_j)$, and the shaded wedge $\mathrm{slack}_i(\varepsilon_i, \varepsilon_j) = \bar{c}(\varepsilon_i) - A_i(\varepsilon_i, \varepsilon_j) \ge 0$ between them.

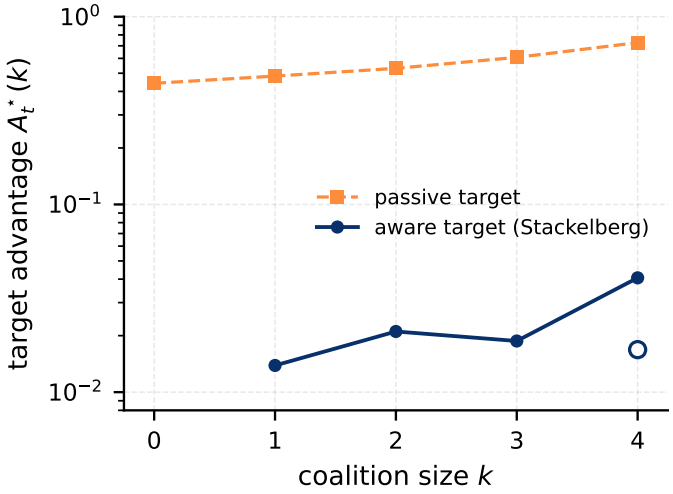


Fig. 4: **H4 — The C4 attack collapses against an aware target.** Target's realised MIA advantage $A_t^\star$ vs coalition size k (log-y; larger is a worse attack for the target). *Orange squares (dashed):* passive target pinned at ε_t=32 as in Fig. 3; A_t climbs monotonically from 0.442 at k=0 (no coalition) to 0.728 at k=4, recovering the empirical budget-manipulation attack of [8]. *Blue circles (solid):* aware target best-responding on the same 14-point log-spaced grid $\varepsilon \in [0.5, 64]$; $A_t^\star \in \{0.014, 0.021, 0.019, 0.041\}$ for $k \in \{1, 2, 3, 4\}$, i.e. 17.9× to 34.8× below the passive curve and ∼11× below even the k=0 baseline. The aware coalition's Stackelberg budgets are $\varepsilon_C^\star \in [30, 64]$ for $k \leq 3$ with the target retreating to $\varepsilon_t^\star \in [9.9, 14.4]$. At k=4 the leader problem has two local optima: the filled blue marker is the Stackelberg argmax, a boundary *poisoning corner* at the grid floor (ε_C=0.5, $\varepsilon_t^\star$=0.73, σ=2.4) that yields the coalition itself only A_C=0.025; the hollow blue marker is the *interior* local optimum (ε_C=30.3, A_t=0.017), which is continuous with $k \leq 3$ and is what a rational coalition would play. Under either k=4 interpretation the collapse is ≥ 17×. Scenario otherwise matches Fig. 3: n=5 symmetric aware players, $|G_i|$=5000, E_b=128, 5 epochs, $\delta = 10^{-12}$.

The wedge grows smoothly from 0.79 at $\varepsilon_j = 0.5$ to 0.96 at $\varepsilon_j = 64$. The DP guarantee correctly upper-bounds the risk—the post-processing-tight inequality $A_i \leq \bar{c}(\varepsilon_i)$ holds on every feasible cell of the grid—but it overpromises by nearly the entire adversarial channel once the other player's budget is loose. The (ε_i, δ) pair alone does not reveal this gap; it is a property of the whole profile. This is the numerical content of the slack decomposition and the visual anchor for the Δ-iDP contract of [8], [11], which would cap this wedge from below by requiring slack_i not to differ too much from a reference at a strategic profile.

H. C6 — Where does the NE locus live in $(\varepsilon_1, \varepsilon_2)$ space?

The eight pure NEs of C1 sit at discrete grid points; the underlying question is whether they are an artefact of the grid resolution or whether they reflect a smaller stable set of isolated equilibria of the continuous game. The implicit first-order condition (16) defines two BR loci—one per player, each a one-dimensional curve in $(\varepsilon_1, \varepsilon_2)$-space—whose intersections *are* the pure NEs. Under symmetric players the two loci are reflections across the diagonal. The number of intersections depends on how the curves bend through each other, and need not grow with the grid resolution.

We re-run C1 at $K = 44$, doubling the resolution along each axis from $K = 22$ (so the grid is now four times denser). Fig. 7 reports the resulting BR loci with the pure NEs marked. The BR curves are clearly visible as one-dimensional stepped loci, and they cross at *eight* distinct points—the same count as at $K = 22$. The crossings sit at nearly identical $(\varepsilon_1, \varepsilon_2)$ values to those at the coarse resolution, modulo grid quantisation. The asymmetric pure NEs are therefore a finite set of isolated equilibria of the continuous game, not a one-parameter family; the discrete grid samples them faithfully.

I. C7 — How does the NE locus respond to the players' privacy weight?

The implicit-curve characterisation of (16) says the BR is parametrised by the privacy weight w_i; we now sweep this weight *symmetrically* across both players from $w = 0.125$ (utility-heavy) up to $w = 4$ (privacy-heavy) and read off how the NE locus moves. Figure 8 reports the union of pure NEs across all six values on a single plot.

The number of pure NEs and their locations vary systematically with w. At the utility-heavy end $w = 0.125$, only a single asymmetric pair of NEs survives, at $(10.08, 64)$ and its swap: the externality is negligible relative to the model-utility gain of a large ε, so both players push toward the upper boundary. At $w = 0.25$ and $w = 0.5$ the NE set is a cluster of six asymmetric NEs; at $w = 1$ we recover the eight of C1; at $w = 2$ the count drops to three, one of them the diagonal profile $(1, 1)$; and at the privacy-heavy end $w = 4$ only a single pure NE remains, on the diagonal at $(\varepsilon^{\min}, \varepsilon^{\min}) = (0.5, 0.5)$—both players race to the strongest privacy the participation constraint admits.

Two readings of this. First, the asymmetric-NE phenomenon of C1 is not fragile: it persists across roughly an order of magnitude of w (roughly $w \in [0.125, 1]$). Second, at the privacy-heavy end the asymmetric NEs disappear and are replaced by a diagonal race to $\varepsilon^{\min}$. In short, the intermediate regime is the one in which the externality matters strategically; the extreme privacy-heavy regime washes it out into a symmetric race to the participation floor.

J. C8 — What happens under asymmetric privacy preferences?

C7 sweeps the privacy weight symmetrically across both players. We now break the preference symmetry: we fix $w_2 = 1$ and vary $w_1 \in \{0.25, 0.5, 1, 2, 4\}$. Player 1 is therefore four times more privacy-sensitive than player 2 at one extreme and four times less sensitive at the other. Figure 9 plots the resulting NEs.

The NE locus shifts smoothly. When $w_1 \ll w_2$ (player 1 values privacy less), pure NEs sit in the lower-right region of the action plane—player 1 plays high ε (she does not mind the realised advantage) and player 2 plays low ε (she does).

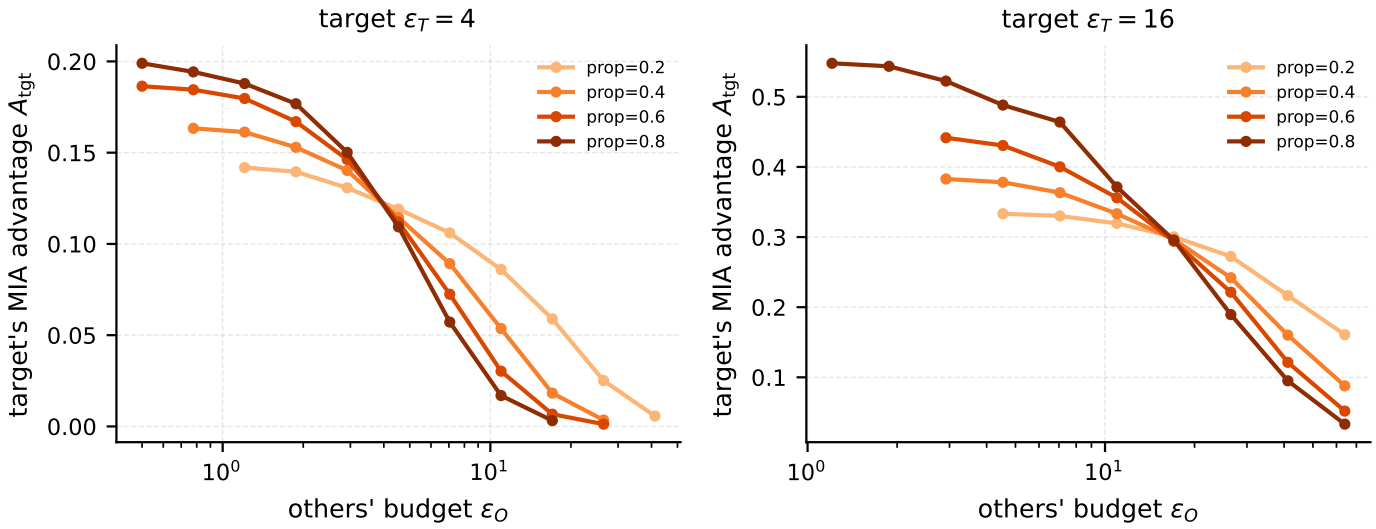


Fig. 5: **C5.** Target's realised MIA advantage as a function of the other group's budget ε_O and proportion, at two fixed target budgets $\varepsilon_T \in \{4, 16\}$. The curves cross at $\varepsilon_O = \varepsilon_T$. Below crossover, larger proportions raise A_t; above crossover, larger proportions lower it.

As w_1/w_2 crosses unity the NEs move through the symmetric configuration of Section VI-B and continue past it: at $w_1 \gg w_2$ the locus has flipped to the upper-left region, with player 1 now in the low-ε role. The structure of the externality therefore makes the *more* privacy-sensitive player carry the lower ε at the equilibrium and the realised advantage lands on the less-sensitive player.

This is unintuitive at first reading but is precisely what the externality lemma predicts. A player who values privacy more is willing to give up more model utility to suppress her own A; doing so pushes σ upward and the other player's p upward, leaving the less-sensitive player to absorb the realised advantage. Critically, asymmetric *preferences* amplify the equilibrium asymmetry rather than dampening it: a population with heterogeneous privacy weights does not converge to a compromise on the diagonal but instead splits, with each player settling into the role for which her weight is best suited. Heterogeneity, in short, does not average out; it self-sorts.

K. H3 — Does the asymmetric NE survive at $n > 2$?

C1 and C8 pin the asymmetric-NE and self-sorting phenomena at $n = 2$; the natural reviewer question is whether either survives once the game scales past a duopoly. We sweep $n \in \{3, 4, 5, 6, 7, 8, 10\}$ on the $K = 7$ log-grid $\varepsilon \in \{1, 2, 4, 8, 16, 32, 64\}$: at $n \leq 5$ we brute-force enumerate the pure NE set, and at $n \in \{6, 7, 8, 10\}$ we run 32 seeds of Monte-Carlo fictitious play (400 rounds; best response against 64 samples of the empirical joint) and cluster the last-50-round FP tail. Both phenomena extend to every n probed, corroborating the small-p existence argument of Theorem 12 at $n > 2$ under strategic complementarities in the shared σ channel.

Under heterogeneous $w_i \in \text{geomspace}(0.25, 4)$, the sort-order compliance rate $\Pr[\tau(w_i, \varepsilon_i) < 0]$ is exactly 1.0 at every $n \in \{3, 4, 5, 6, 7, 8\}$ and 0.83 at $n = 10$ (the residual 17% at $n = 10$ reflects FP-tail excursions that swap adjacent w-close player rankings without flipping the sign of the sorting monotonicity); the $A_{\max}/A_{\min}$ envelope is bounded away from unity throughout, running from 483 at $n = 3$ up to 637 at $n = 4$ and then decaying monotonically to 75 at $n = 10$ as the FP tail averages more players (Fig. 10). Under symmetric w, the $K = 7$ grid pins the FP tail to the diagonal for $n \leq 8$—a discretisation limit, as the no-symmetric-NE claim rides on the continuous-ε argument of Section VI-B, not on the discrete grid; at $n = 10$ FP additionally explores strict-asymmetric profiles with a modest envelope ratio of 4.8. The $n = 2$ self-sorting proposition therefore extends empirically to n up to 10, consistent with the supermodular existence result.

L. C9 — Price of externality-vulnerability at the equilibria

The equilibria of Sections VI-B, VI-I and VI-J are welfare-costly in a quantifiable sense. We measure that cost with a value-side analogue of the Price of Anarchy: the relative inflation of aggregate realised MIA advantage at a reference profile against a symmetric uniform-budget baseline.

Definition 17 (Price of externality-vulnerability)**.** Given a symmetric baseline profile $\boldsymbol{\varepsilon}^{\text{uni}} \in \mathcal{A}_K^n$ and a reference profile $\boldsymbol{\varepsilon}^\star$, the *price of externality-vulnerability* at $\boldsymbol{\varepsilon}^\star$ is

$$\text{PoEV}(\boldsymbol{\varepsilon}^\star) := \frac{\sum_i A_i(\boldsymbol{\varepsilon}^\star) - \sum_i A_i(\boldsymbol{\varepsilon}^{\text{uni}})}{\sum_i A_i(\boldsymbol{\varepsilon}^{\text{uni}})}.$$

PoEV > 0 signals that the reference profile inflates aggregate advantage relative to the baseline; PoEV < 0 signals deflation (either the utility term pulls both players to a modest-noise profile with modest A, or a redistributive NE concentrates advantage on one player while sparing the other).

Table II reports PoEV at each pure NE and at the social optimum of the symmetric two-player game on the $K = 200$

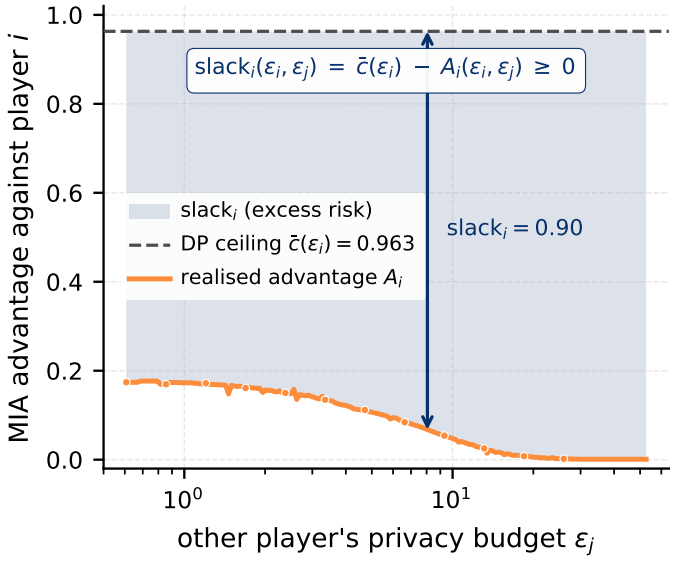


Fig. 6: **Excess risk (slack) uncounted by the DP guarantee at anchor** $\varepsilon_i = 4$**.** The dashed grey line is the post-processing-tight (ε_i, δ)-DP upper bound on the MIA advantage, $\bar{c}(\varepsilon_i) = (e^{\varepsilon_i} - 1 + 2\delta)/(e^{\varepsilon_i} + 1)$; the orange curve is the realised MIA advantage $A_i(\varepsilon_i, \varepsilon_j)$ of the two-fold subsampled Gaussian at profile $(\varepsilon_i, \varepsilon_j)$, computed on the $K = 200$ mechanism grid at $|G_1| = |G_2| = 5000$, $E_b = 128$; the shaded wedge is $\text{slack}_i(\varepsilon_i, \varepsilon_j) = \bar{c}(\varepsilon_i) - A_i(\varepsilon_i, \varepsilon_j) \geq 0$. As the other player's budget loosens (larger ε_j), the realised advantage collapses toward zero and the slack fills almost the entire adversarial channel: the guarantee correctly upper-bounds the risk but overpromises by a factor that the (ε, δ) pair alone does not reveal. Compare with Fig. 5: the two figures share the same monotone envelope, viewed from below (this figure, as slack from the ceiling) versus from above (Fig. 5, as absolute advantage).

grid, against the geometric-midpoint uniform baseline $\varepsilon^{\text{uni}} = 5.73$ with $\sum_i A_i(\varepsilon^{\text{uni}}) = 0.305$. At the canonical $w_1 = w_2 = 1$ setting almost every NE is deflationary: the median PoEV across the 48 pure NEs resolved at $K = 200$ is -0.50 and the maximum is $+0.04$ at the most-asymmetric NEs $(1.43, 9.33)$ and its reflection. The social optimum inverts the naive PoA expectation: it consolidates both players on a moderate-ε, low-A profile at $(10.28, 12.80)$ and reduces aggregate advantage by 78% ($\text{PoEV}_{\text{SO}} = -0.78$).

Reading this against the C7 sweep, PoEV moves monotonically with the privacy weight. At $w = 0.125$ the value term dominates, NEs push into the high-ε regime, and 77% of them have PoEV > 0 with a maximum of $+1.70$ at $(9.56, 64.00)$; at the privacy-heavy end $w = 4$ every NE has PoEV ≤ -0.76, the game having collapsed to the participation floor. C8's asymmetric-weight sweep shows the same pattern with a different driver: when player 1 is $4\times$ less privacy-sensitive than player 2, she pushes ε_1 toward the grid top and floods player 2 with realised advantage, producing PoEV up to $+1.67$. The numerical range of PoEV across all $(K = 200)$ sweeps is $[-0.85, +1.70]$; any subsequent analytic bound on the aggregate externality has to respect these values.

TABLE II: Numerical price of externality-vulnerability (PoEV, Theorem 17) at each pure Nash equilibrium of the symmetric 2-player PEG and at its social optimum. $K = 200$ log-spaced action grid on $[0.5, 64]$, $|G_1| = |G_2| = 5000$, $E_b = 128$, 5 epochs, $\delta = 10^{-12}$, $w_1 = w_2 = 1$, $V(\sigma) = 1/(1 + \sigma^2)$. Uniform baseline $\varepsilon^{\text{uni}} = 5.73$ (geometric midpoint of the log grid) with $\sum_i A_i(\varepsilon^{\text{uni}}) = 0.305$. Social optimum is the feasible action profile maximising $u_1 + u_2$; it consolidates both players on a moderate-ε, low-A profile and is the most deflationary point in the table.

	$(\varepsilon_1^*, \varepsilon_2^*)$	A_1	A_2	$\sum_i A_i$	PoEV
NE_1	$(1.43, 9.33)$	0.001	0.295	0.296	-0.032
NE_2	$(1.53, 3.88)$	0.011	0.142	0.153	-0.500
NE_3	$(1.91, 3.35)$	0.030	0.117	0.146	-0.521
NE_4	$(3.35, 1.91)$	0.117	0.030	0.146	-0.521
NE_5	$(3.88, 1.53)$	0.142	0.011	0.153	-0.500
NE_6	$(4.07, 4.17)$	0.055	0.059	0.114	-0.626
NE_7	$(4.17, 4.07)$	0.059	0.055	0.114	-0.626
NE_8	$(9.33, 1.43)$	0.295	0.001	0.296	-0.032
social opt	$(10.28, 12.80)$	0.017	0.048	0.066	-0.785

VII. Discussion and Conclusion

We have shown that sampling-based individual differential privacy is not just *vulnerable* to collusion attacks but *equilibrium-asymmetric*. The pure Nash equilibria of the Privacy Externality Game, when they exist on the dense action grid, sit *off the diagonal*: one player accepts a higher realised MIA advantage and is compensated by a higher model-utility benefit; the other absorbs the opposite trade. The budget-manipulation attack of [8] is precisely the asymmetric equilibrium played by a coalition that weights the target's harm in its joint utility, and Theorem 14 shows this is the rational best response of *any* coordinated subset of data holders.

Limitations of this work. Three caveats are worth noting. First, our model assumes one-shot full-information play. Real deployments of iDP are dynamic: contributors join and leave; budgets are re-negotiated; trainers re-train. A natural extension is a repeated game in which the players also have private preference types and learn about one another across rounds. Section VII-A formalises such a game as the *Bayesian Repeated PEG* (BR-PEG), states a sharpened attack characterisation (Theorem 18) that identifies the optimal adversarial budget as strictly interior to the feasible action grid, and reports a small simulation. We treat the fully two-sided repeated game of [37] and quantitative regret rates in the style of [38] as follow-up work. Second, we assume full information about other players' actions. Real contracts may keep budgets private, and Theorem 6 already distinguishes between fully aware players and players who only see their nominal (ε_i, δ) contract. A Bayesian PEG that treats ε_i as a private type drawn from a known distribution is a natural and important extension. Third, our analysis is restricted to the sampling-based realisation of iDP. The sensitivity-based realisation [2] adjusts per-sample

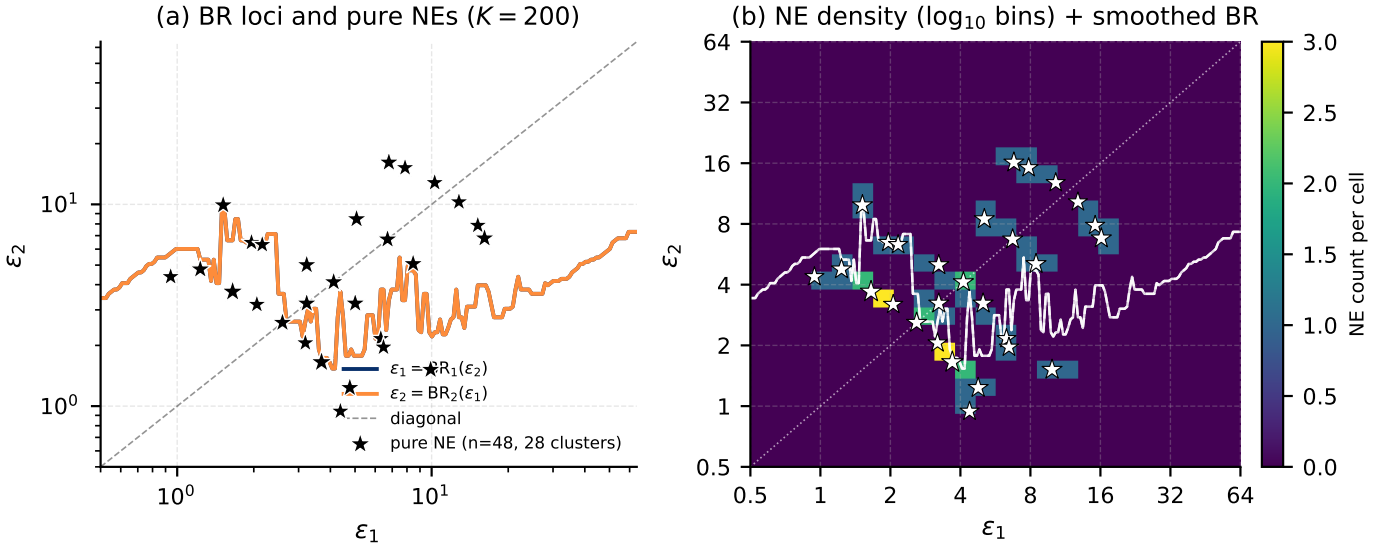


Fig. 7: **C6.** BR loci and pure NEs (left) and the u_1 surface with NEs marked (right) at the refined resolution $K = 44$. The two BR correspondences are one-dimensional stepped curves in the action plane; they cross at exactly eight isolated points (the same count as at $K = 22$), confirming that the asymmetric NE set is finite.

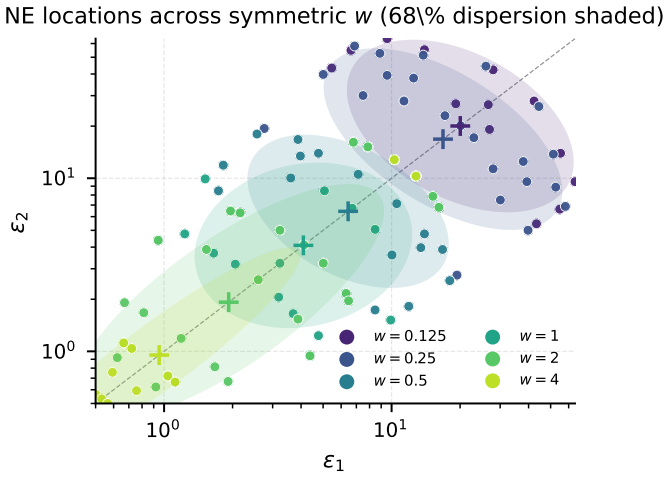


Fig. 8: **C7.** Pure NE locations across symmetric privacy weights $w \in \{0.125, 0.25, 0.5, 1, 2, 4\}$ at $K = 22$. The dashed diagonal is for reference. NEs are asymmetric for $w \in [0.25, 2]$ and collapse to the diagonal at the utility-heavy and privacy-heavy extremes.

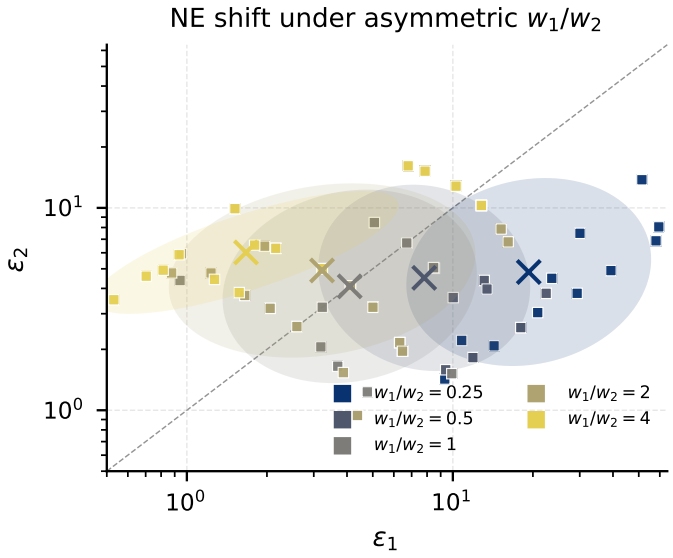


Fig. 9: **C8.** Pure NE locations under asymmetric privacy weights $w_1/w_2 \in \{0.25, 0.5, 1, 2, 4\}$ at $K = 22$ (with $w_2 = 1$). As w_1/w_2 grows (player 1 more privacy-sensitive), the NE cloud migrates from the lower-right region to the upper-left region of the action plane.

clipping bounds rather than per-sample sampling rates and does not exhibit the same externality (the noise-to-sensitivity ratio is preserved by construction). Our framework specialises to that case as the trivial PEG in which the externality vanishes.

A. Repeated Bayesian variant (BR-PEG)

The one-shot analysis leaves the strategic content of repetition and incomplete information untouched. We formalise the natural extension— the *Bayesian Repeated PEG*—and identify the qualitative consequence of the mechanism's leader–follower coupling: the optimal adversarial budget is bounded away from the minimum feasible one.

The game. Two players, H (honest) and M (malicious). At $t = 0$, H draws a private type $\theta_H = (w_H, n_H)$ from a finite set Θ known to M. Per round $t \in \{1, \ldots, T\}$: (i) M observes the history and picks (ε_M^t, n_M^t); (ii) H observes (ε_M^t, n_M^t) and myopically best-responds with $\varepsilon_H^t \in \mathcal{A}$ that maximises the one-shot utility $u_H(\varepsilon_H; \varepsilon_M^t, \theta_H) = V(\sigma) -$

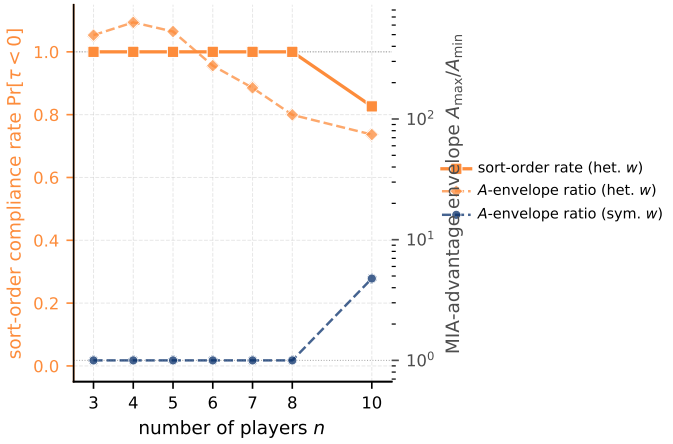


Fig. 10: **H3.** Two summary diagnostics of the n-player Privacy Externality Game as a function of the player count n on the $K = 7$ log-grid $\varepsilon \in \{1, 2, 4, 8, 16, 32, 64\}$: pure NEs at $n \le 5$ (brute-force enumeration) and the last-50-round tail of Monte-Carlo fictitious play at $n \in \{6, 7, 8, 10\}$ (32 random-init seeds $\times$ 400 rounds; best response against 64 samples of the empirical joint). *Left axis:* sort-order compliance rate—the fraction of observations with Kendall $\tau(w_i, \varepsilon_i) < 0$ under heterogeneous $w_i \in \text{geomspace}(0.25, 4)$; a rate near 1 is the self-sorting signature of Section VI-J. *Right axis (log):* the $A_{\max}/A_{\min}$ envelope ratio of realised MIA advantage across every observed profile, for both regimes; a ratio bounded away from 1 is the strict-asymmetric-equilibrium signature of Section VI-B. Both signatures survive throughout $n \in \{3, \dots, 10\}$ in the heterogeneous regime; the symmetric-w envelope pinned at 1 for $n \le 8$ is a $K = 7$ discretisation artefact, resolved by the continuous-ε argument in Section VI-B.

$w_H A_H$, with $(\sigma, p_H, p_M, A_H, A_M)$ returned by the sampling-based iDP mechanism of Theorem 1; (iii) the pair (ε_H^t, u_H^t) becomes public and M updates a posterior $\pi^t \in \Delta(\Theta)$ over θ_H via the observation likelihood induced by H's best-response map. M's total score is $\sum_t A_H^t$. Positioned in the literature, the game is the Stackelberg-learning template of [38]–[40] with the tabular follower payoff replaced by the DP-accountant coupling of Theorem 1, and it is the myopic-informed special case of the one-sided incomplete-information repeated game of [37]. The prior π^0 has full support on Θ, so Kalai–Lehrer's absolute-continuity condition holds trivially and posterior consistency is automatic; the interesting content is what M does once consistency kicks in.

Proposition 18 (Bayesian-greedy attack is not min-budget)**.** *Consider the BR-PEG with H myopic and M posterior-greedy on a finite action grid $\mathcal{A} \subset [\varepsilon^{\min}, \varepsilon^{\max}]$. Fix a type $\theta_H^\star \in \Theta$ and let*

$$G(\varepsilon_M; \theta) \;:=\; A_H\big(\varepsilon_M,\, \varepsilon_H^{\mathrm{BR}}(\varepsilon_M; \theta)\big)$$

be the composed map from M's action to the elicited advantage, where $\varepsilon_H^{\mathrm{BR}}(\varepsilon_M; \theta) \in \arg\max_{\varepsilon_H \in \mathcal{A}} u_H(\varepsilon_H; \varepsilon_M, \theta)$. Assume

(I) Identifiability. *The observation likelihood is $\theta_H^\star$-identifying on the equilibrium path in the sense of [41]: for a full-support prior π^0, $\pi^t(\theta_H^\star) \to 1$ π^0-a.s.*

(A) Interior argmax. *$G(\cdot; \theta_H^\star)$ attains its maximum on $\mathcal{A}$ at a unique $\varepsilon_M^\star(\theta_H^\star) \in \mathcal{A}$ with $\varepsilon_M^\star(\theta_H^\star) > \varepsilon^{\min}$.*

Then the posterior-greedy action $\varepsilon_M^{\mathrm{greedy}}(t) \in \arg\max_{\varepsilon_M \in \mathcal{A}} \mathbb{E}_{\theta \sim \pi^t}\big[G(\varepsilon_M; \theta)\big]$ satisfies $\varepsilon_M^{\mathrm{greedy}}(t) = \varepsilon_M^\star(\theta_H^\star)$ eventually, π^0-a.s.; equivalently, setting $\eta := \varepsilon_M^\star(\theta_H^\star) - \varepsilon^{\min} > 0$, $\varepsilon_M^{\mathrm{greedy}}(t) \ge \varepsilon^{\min} + \eta$ for all sufficiently large t.

Proof sketch. By (I), $\pi^t \to \delta_{\theta_H^\star}$ in total variation. Since G is bounded ($0 \le A_H \le 1$), the posterior-averaged score $\bar{G}(\varepsilon_M; \pi^t) := \mathbb{E}_{\theta \sim \pi^t} G(\varepsilon_M; \theta)$ satisfies $|\bar{G}(\varepsilon_M; \pi^t) - G(\varepsilon_M; \theta_H^\star)| \le 1 - \pi^t(\theta_H^\star) \to 0$, uniformly over the finite grid $\mathcal{A}$. By (A), the truth-argmax gap $\Delta := G(\varepsilon_M^\star; \theta_H^\star) - \max_{\varepsilon_M \ne \varepsilon_M^\star} G(\varepsilon_M; \theta_H^\star) > 0$; once $1 - \pi^t(\theta_H^\star) < \Delta/2$, the argmax of $\bar{G}(\cdot; \pi^t)$ coincides with $\varepsilon_M^\star(\theta_H^\star)$ for every subsequent round. Section F works the argument through in full. □

Numerical realisation and scope. For the demo instance of Section F ($\Theta = \{w_H \in \{0.5, 1, 2\}\}$, $n_H = n_M = 5000$, H's true type $w_H^\star = 2$), the posterior collapses at $t = 1$, and from then on the Bayesian-greedy attack plays $\varepsilon_M^\star \approx 3.08$ and elicits $A_H \approx 0.454$—roughly twice the naive-minimum attack $\varepsilon_M = 1$ and fifty times the benign baseline $\varepsilon_M = 64$. The separation from the naive-attack anchor is $\eta \approx 2.08$; assumption (A) is verified numerically on the mechanism grid. Two extensions sit outside this scope. First, the strategising-H case, in which H chooses her action sequence to manage what she reveals: a two-sided incomplete-information game in the sense of [37]. Second, a regret rate in the style of [38], [40] that quantifies the number of rounds until $\varepsilon_M^{\mathrm{greedy}}(t) = \varepsilon_M^\star$; the finite-type Kalai–Lehrer setup used here delivers one-round identification and so provides only a constant-regret upper bound. Both extensions fit the framework above without modification.

B. Modelling choices, mitigation, and open questions

The role of the model-utility function V. Our choice $V(\sigma) = 1/(1 + \sigma^2)$ is motivated by the convex DP-ERM excess-risk bound of [10], and we use it as a directional proxy in our possibly non-convex setting. The qualitative findings—existence of asymmetric NEs, monotone coalition attack, externality lemma—do not depend on the precise shape of V; they require only that V is continuous, monotone-decreasing in σ, and bounded. We expect the same qualitative picture under any utility function with these three properties.

Mitigation. The natural mitigation is to make the contract internalise the externality. The $(\varepsilon_i, \delta, \Delta)$-iDP formulation of [11] adds a Δ-divergence bound on the realised vulnerability, restricting the action space of the PEG to profiles that satisfy a per-record bound on excess advantage. Theorem 19 makes the deterministic half of this statement precise: capping the slack at Δ squeezes every player's realised advantage into $[\bar{c}(\varepsilon_i) - \Delta, \bar{c}(\varepsilon_i)]$ and caps its cross-profile variation at fixed ε_i

by Δ. Existence of a unique equilibrium under this restriction is left to future work (the second open question below).

Proposition 19 (Δ-neutralisation of the externality)**.** *Restrict the action set to profiles on which* $\mathrm{slack}_i(\boldsymbol{\varepsilon}) \le \Delta$ *for every player* i *(equivalently: the* $(\varepsilon_i, \delta, \Delta)$*-iDP contract of [11] is enforced by the mechanism). Then:*

(i) Pointwise squeeze. *At every feasible profile* $\boldsymbol{\varepsilon}$ *in the restricted action set,*

$$\bar{c}(\varepsilon_i) - \Delta \;\le\; A_i(\boldsymbol{\varepsilon}) \;\le\; \bar{c}(\varepsilon_i).$$

(ii) Cross-profile bound at fixed own budget. *For any two feasible profiles* $\boldsymbol{\varepsilon}, \boldsymbol{\varepsilon}'$ *in the restricted action set that agree in* i*'s own coordinate* ($\varepsilon_i = \varepsilon_i'$)*,*

$$\left|A_i(\boldsymbol{\varepsilon}) - A_i(\boldsymbol{\varepsilon}')\right| \;\le\; \Delta.$$

That is: once the slack is capped by Δ*, no unilateral or joint deviation of the other players can move player* i*'s realised advantage by more than* Δ *from any other feasible value it can attain at the same* ε_i*; and* A_i *never falls below its DP worst case by more than* Δ*.*

Proof. Both parts use only the identity (5) and the non-negativity of slack_i.

(i) Pointwise squeeze. By definition $\mathrm{slack}_i(\boldsymbol{\varepsilon}) \ge 0$, so $A_i(\boldsymbol{\varepsilon}) = \bar{c}(\varepsilon_i) - \mathrm{slack}_i(\boldsymbol{\varepsilon}) \le \bar{c}(\varepsilon_i)$. The restriction $\mathrm{slack}_i \le \Delta$ then gives $A_i(\boldsymbol{\varepsilon}) \ge \bar{c}(\varepsilon_i) - \Delta$.

(ii) Cross-profile bound. For two profiles with $\varepsilon_i = \varepsilon_i'$, $\bar{c}(\varepsilon_i)$ cancels and

$$A_i(\boldsymbol{\varepsilon}) - A_i(\boldsymbol{\varepsilon}') \;=\; \mathrm{slack}_i(\boldsymbol{\varepsilon}') - \mathrm{slack}_i(\boldsymbol{\varepsilon}).$$

Both slacks lie in $[0, \Delta]$ by the restriction and by non-negativity, hence the difference lies in $[-\Delta, \Delta]$ and $|A_i(\boldsymbol{\varepsilon}) - A_i(\boldsymbol{\varepsilon}')| \le \Delta$. □

Open questions. The following questions naturally arise from this work but lie outside its scope:

1) Does a pure-strategy NE always survive on the continuous action space under the coupled cost, and does a Δ-bound on the slack suffice for existence?
2) Is the PEG a potential game in the sense of [32]? PoP's analysis was a potential game because the privacy cost was separable; the externality term breaks this directly. Conditions on the cross-partial of A_i that recover the potential structure would unify the two settings.
3) What is the sign of the externality across composition regimes (minority vs majority weak-privacy)? Our experiments suggest the smaller group is favoured, but the analytic dependence on $|G_g|$ deserves a closed-form characterisation.
4) What is the appropriate player granularity (single sample versus silo) in the FL setting, and how does the granularity change the equilibria?
5) Worst-case (analytical) vs realised (empirical) cost—which should the contract be written against, and how large is the gap in practice across deep-learning workloads?
6) Is the $(\varepsilon_i, \delta, \Delta)$-iDP contract by itself sufficient as a collusion-resistant mechanism, or are sensitivity-based iDP or public-data padding needed when the slack is exceeded?

Wider lesson. The wider lesson is that an *individual* privacy contract is incomplete: it does not pin down the collective dynamics. Where individuals interact strategically (in a single machine-learning training, in a federated round, in any future composition of the iDP mechanism), the contract must also constrain their joint behaviour or the equilibrium will quietly violate at least one participant's expectations while preserving every formal guarantee.

Conclusion. The promise of iDP—that every data contributor controls her own privacy—is broken by the privacy externality of sampling-based mechanisms. We formalised the strategic situation faced by iDP data contributors as the Privacy Externality Game, proved monotonicity of the externality and characterised the equilibria. The result is that the collusion attack on iDP is not adversarial but *rational*, and the remedy must operate at the level of the contract.

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APPENDIX A
MECHANISM-SOLVER CONSTRUCTION

The mechanism solution $(\sigma, p_1, \ldots, p_n)$ for a given action profile $\boldsymbol{\varepsilon}$ is obtained by a nested bisection. For each candidate σ, we find the largest per-group p_g that satisfies the I-fold (p_g, σ)-subsampled Gaussian at (ε_g, δ)-DP under (C1). Because the achieved ε is monotone-increasing in p_g, a unique $p_g \in (0, 1]$ exists (capped at 1 if the budget cannot be exhausted); we locate it by binary search using the Rényi-DP accountant [34] until the achieved ε matches ε_g to a relative tolerance of 10^{-3}. We then bisect σ on top of this inner search, using the realised batch size $B(\sigma) = \sum_g |G_g|\, p_g(\sigma, \boldsymbol{\varepsilon})$ as the outer-loop monitor, until B matches E_b to the same tolerance. Profiles in which any p_g falls below 10^{-5} are flagged infeasible per Theorem 4; below that threshold the group's contribution to (C2) is numerically negligible and the iDP contract is trivially satisfied. The qualitative results are insensitive to this threshold across $[10^{-6}, 10^{-3}]$.

Once (σ, p_g) is fixed, the realised MIA advantage of group g is computed exactly from the privacy-loss-distribution of the I-fold (p_g, σ)-subsampled Gaussian [28] by evaluating the trade-off function and (2). This matches the formulation of [8] and yields the tightest analytical MIA advantage for the resulting mechanism.

APPENDIX B
VALIDITY OF $V(\sigma) = 1/(1+\sigma^2)$

[10] prove that for a 1-Lipschitz strongly-convex $\ell\colon \mathcal{X} \times \mathcal{D} \to \mathbb{R}$, the excess empirical risk of DP-ERM satisfies

$$\mathbb{E}[\hat{L}(\widehat{\theta}) - \hat{L}(\theta^*)] \;=\; O\!\left(\frac{d\,\log(1/\delta)}{n^2\,\varepsilon^2}\right).$$

Under the moments accountant of [26] this translates into a $\sigma^2 \cdot d/n^2$ dependence of the excess risk. Treating model utility as the negative of excess risk and renormalising to $[0, 1]$ gives $V(\sigma) \propto 1/(1 + \alpha\,\sigma^2)$ for a problem-dependent constant α; we set $\alpha = 1$ for simplicity. The bound assumes convexity, but the empirical literature on DP-SGD in deep learning [2], [26], [27] confirms the same qualitative direction. The equilibrium results that follow do not depend on the particular functional form: only on V being continuous, monotone-decreasing in σ, and bounded. This informal statement is formalised as Theorem 16 in Section V, whose proof inspects the proofs of Theorem 7 and Theorem 14 and verifies that V enters only through $\operatorname{sgn} V'$.

APPENDIX C
SENSITIVITY CHECKS

We re-ran the 2-player setting of Fig. 1 at $K = 44$ (twice the resolution along each axis) and obtained the same eight pure NEs at nearly identical locations; the asymmetric multiplicity is therefore not a grid artefact. Substituting $V(\sigma) = e^{-\sigma}$ for $V(\sigma) = 1/(1 + \sigma^2)$ shifts the NEs quantitatively—they sit at somewhat higher ε values because the exponential decays faster as σ grows—but the asymmetric structure is preserved (consistent with Theorem 16). Lowering the participation threshold from 10^{-5} to 10^{-6} enlarges the feasible region and uncovers a few additional NEs on the new interior; raising it to 10^{-3} removes some boundary NEs while leaving the interior ones unchanged.

A. Robustness to the utility approximation V

The main body fixes a single shared proxy $V_1(\sigma) = 1/(1+\sigma^2)$ and derives every equilibrium sweep against it. The equilibrium results of Section V depend on V only through the requirement that it be continuous, strictly monotone-decreasing on the feasible σ-range, and bounded on that range. Here we instantiate that observation numerically on the same $K = 200$ mechanism grid used by Section VI-H at symmetric weights $w_1 = w_2 = 1$, sweeping six qualitatively distinct admissible shapes: the canonical rational $V_1(\sigma) = 1/(1+\sigma^2)$, the slower-decaying $V_2(\sigma) = 1/(1+\sigma)$, the Gaussian $V_3(\sigma) = e^{-\sigma^2/2}$, the exponential $V_4(\sigma) = e^{-\sigma}$, the concave $V_5(\sigma) = -\log(1+\sigma)$, and the affine $V_6(\sigma) = -\sigma$. Because the mechanism grid is V-free (utility (3) separates additively into the shared $V(\sigma)$ and the private $w_i A_i$), each V requires only $O(K^2)$ elementwise operations on the cached mechanism tensor.

Across the six shapes the pure-NE count on the fixed $K = 200$ discretisation stays in the range 44–60, and the entire NE set remains off-diagonal for every V: the qualitative asymmetry diagnosed in Section VI-B is not an artefact of the canonical proxy. The one-sided Hausdorff distance from each V_1-NE to its nearest NE under V_k has mean at most 0.11 decades of $\log_{10}\varepsilon$, so the baseline equilibrium locus sits inside a small tubular neighbourhood of the perturbed loci. Externality-sign agreement with V_1 is 100% over the interior feasible edge set for every V_k, and the per-V externality-sign fraction itself is 82.19725% identical to five decimal places—the numerical fingerprint of the additive separation of (3), in which the sign of $\partial u_i/\partial \varepsilon_j$ is mechanism-determined and V-free on the feasible interior.

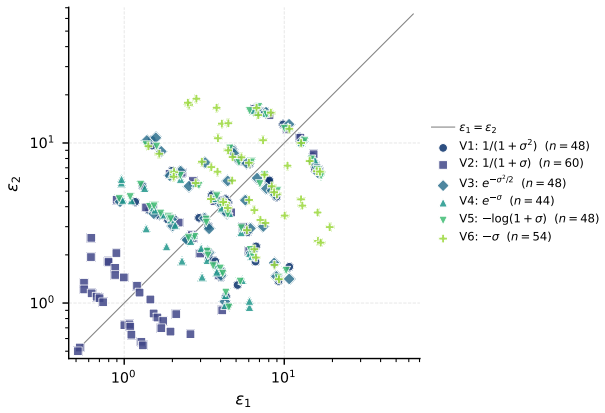


Fig. 11: **Utility-approximation ablation.** Pure NE locations at $w_1 = w_2 = 1$ on the $K = 200$ mechanism grid, under six admissible shapes for V: rational $V_1(\sigma) = 1/(1+\sigma^2)$ (canonical), slow-rational $V_2(\sigma) = 1/(1+\sigma)$, Gaussian $V_3(\sigma) = e^{-\sigma^2/2}$, exponential $V_4(\sigma) = e^{-\sigma}$, concave $V_5(\sigma) = -\log(1+\sigma)$, and affine $V_6(\sigma) = -\sigma$. Grey line: the diagonal $\varepsilon_1 = \varepsilon_2$. Under every V, 100% of the pure NEs sit off the diagonal, and the externality-sign agreement rate with the baseline V_1 is 100% over the interior feasible edge set; the qualitative structure diagnosed in Section VI-B is not an artefact of the canonical proxy. The pure-NE count stays in the range 44–60 across shapes, and every V_1-NE has a match within 0.11 decades of $\log_{10}\varepsilon$ under every alternative V.

tag	shape	#NE	mean$(b \to a)$	agree$_{\text{sign}}$
V_1	$1/(1+\sigma^2)$	48	0.000	100.0%
V_2	$1/(1+\sigma)$	60	0.075	100.0%
V_3	$e^{-\sigma^2/2}$	48	0.013	100.0%
V_4	$e^{-\sigma}$	44	0.062	100.0%
V_5	$-\log(1+\sigma)$	48	0.000	100.0%
V_6	$-\sigma$	54	0.111	100.0%

TABLE III: Per-V pure-NE count, mean one-sided Hausdorff distance from a V_1-NE to its nearest NE under V_k (in decades of $\log_{10}\varepsilon$), and externality-sign agreement rate with V_1 over interior feasible edges. $K = 200$, $w = 1$.

APPENDIX D
2-PLAYER SELF-SORTING

The C1 experiment observes that in the asymmetric 2-player PEG the higher-w player carries the smaller ε at every pure NE. In two players this observation is analytic.

Proposition 20 (Self-sorting, 2-player PEG)**.** *Consider the 2-player PEG (Theorem 2) with privacy weights $w_1 > w_2 > 0$, symmetric groups $|G_1| = |G_2|$, and utility $u_i(\boldsymbol{\varepsilon}) = V(\sigma) - w_i\, A_i(\boldsymbol{\varepsilon})$ with V strictly decreasing ($V' < 0$). Write $f(x, y) := A_i(\varepsilon_i = x, \varepsilon_{-i} = y)$; by $|G_1| = |G_2|$ the mechanism's noise satisfies $\sigma(x, y) = \sigma(y, x)$ and f is common to both players.*

Assume constraints (C1) *and* (C2) *bind on the interior of the feasible region and that*

(A) $\partial f/\partial x > 0$ *(own MIA advantage strictly increasing in own budget);*

(S) *for every* $x > y$ *in the feasible region,* $\big[V(\sigma_H) - V(\sigma_M)\big]\big[f(x,y) - f(y,y)\big] \geq \big[V(\sigma_M) - V(\sigma_L)\big]\big[f(x,x) - f(y,x)\big]$*, where* $\sigma_L = \sigma(y,y)$*,* $\sigma_M = \sigma(x,y)$*,* $\sigma_H = \sigma(x,x)$*.*

Then every pure Nash equilibrium $(\varepsilon_1^\star, \varepsilon_2^\star)$ *satisfies* $\varepsilon_1^\star \leq \varepsilon_2^\star$*.*

Proof. Fix a pure NE interior to the feasible region and suppose for contradiction $\varepsilon_1^\star > \varepsilon_2^\star$; write $x = \varepsilon_1^\star$, $y = \varepsilon_2^\star$. By Theorem 7 the noise σ is strictly decreasing in every coordinate of $\boldsymbol{\varepsilon}$, so $\sigma_L > \sigma_M > \sigma_H$; with $V' < 0$, both differences $\Delta V_H := V(\sigma_H) - V(\sigma_M)$ and $\Delta V_L := V(\sigma_M) - V(\sigma_L)$ are strictly positive. By (A) the two "own-budget lifts" $D_1 := f(x,y) - f(y,y)$ and $D_2 := f(x,x) - f(y,x)$ are also strictly positive.

Player 1 has no profitable unilateral deviation from x to y: $V(\sigma_M) - w_1 f(x,y) \geq V(\sigma_L) - w_1 f(y,y)$, i.e. $w_1 D_1 \leq \Delta V_L$. Player 2 has no profitable unilateral deviation from y to x: $V(\sigma_M) - w_2 f(y,x) \geq V(\sigma_H) - w_2 f(x,x)$, i.e. $w_2 D_2 \geq \Delta V_H$. Dividing the second inequality by the first yields

$$\frac{w_2}{w_1} \;\geq\; \frac{\Delta V_H}{\Delta V_L} \cdot \frac{D_1}{D_2}.$$

Assumption (S) is exactly $\Delta V_H \cdot D_1 \geq \Delta V_L \cdot D_2$, i.e. the right-hand side is ≥ 1. Hence $w_2 \geq w_1$, contradicting $w_1 > w_2$. □

Assumption (S) is a two-point exchange inequality that decides whether the swap deviation is more or less attractive to the high-ε player than to the low-ε player. It admits three concrete sufficient sub-conditions: (S1) V midpoint-convex on $[\sigma_H, \sigma_L]$ and f submodular on the $\{y,x\}^2$ block (using (8), (S1) closes when the NE sits in the high-noise regime $\sigma > \sqrt{3}$); (S2) the first-order-condition ratio at any candidate NE evaluates to a value less than 1 for $x > y$ and greater than 1 for $y > x$, which is the differential restatement of (S) via the mean-value theorem; and (S3) point-wise numerical verification at the specific NE candidate on the mechanism grid. On the K=22 mechanism cache the proposition's prediction holds at every strong-asymmetry cell ($|\log_2(w_1/w_2)| \geq 2$) unanimously, and $45/49$ of a 7×7 weight-ratio sweep respects the sort; the four discrepant cells are at mild asymmetry and lie flush against the grid boundary, characteristic signatures of finite-grid discretisation NEs.

At the boundary $w_1 = w_2$, inequality (S) becomes an equality and the proof reduces to the symmetric case: the pure-NE set is closed under the swap $(\varepsilon_1, \varepsilon_2) \leftrightarrow (\varepsilon_2, \varepsilon_1)$, consistent with the paired off-diagonal NEs observed in C1. Extending Theorem 20 to $n \geq 3$ players is the natural next step; in Milgrom–Shannon terms, (A) is the statement that u_i has strictly decreasing differences in (ε_i, w_i), so the monotone comparative-statics apparatus of [42] applies to best-response ordering in w_i, but not directly to the cross-player order $\varepsilon_1 \leq \varepsilon_2$; a supermodularity route in the coupled (ε_i, σ) argument would close that gap.

Appendix E
Non-strategyproofness and a Myerson setup sketch

A natural objection to the strategic-analysis stance of the paper is mechanism-design–flavoured: if the elicitation of budgets ε_i is broken, one could redesign the mechanism instead of analysing equilibria of the broken one. This appendix closes that escape hatch. We show that sampling-based iDP, viewed as a direct-revelation mechanism on the budget channel, is not dominant-strategy incentive-compatible (DSIC): for any player with strictly positive privacy weight, the utility- maximising budget report is a non-constant function of the opponents' declarations, so no single report can be a dominant strategy.

Environment and mechanism. Fix a common $\delta > 0$, an expected batch size E_b, and an iteration count I. Player i's group size $|G_i|$ is public; her type is her privacy weight $\theta_i = w_i \in \mathbb{R}_{>0}$, private to i. A direct-revelation iDP mechanism is a pair (g,t) of an allocation rule and payment rule

$$(g,t)\colon \prod_i \mathbb{R}_{>0} \;\longrightarrow\; \big(\mathbb{R}_{>0} \times [0,1]^n \times \mathbb{R}_{>0}^n\big) \times \mathbb{R}^n,$$

mapping a reported profile $\hat{\boldsymbol{\theta}}$ to a mechanism solution $(\sigma, \{p_i\}, \{\varepsilon_i^{\text{alloc}}\})$ satisfying (C1)–(C2) of Theorem 1 together with a transfer $\{t_i\}$. Player i's utility at true type w_i is $u_i(g(\hat{\boldsymbol{\theta}}); w_i) - t_i(\hat{\boldsymbol{\theta}}) = V(\sigma) - w_i A_i - t_i$. Sampling-based iDP as currently used, and throughout this paper, is the special case in which each player declares her chosen budget $\hat{\varepsilon}_i \in \Sigma_i$ directly, the mechanism solver produces $(\sigma, \{p_j\})$ from that profile, and $t_i \equiv 0$; we call this the *identity mechanism* on the budget channel.

Why Myerson without payments applies here. The classical single-parameter Myerson characterisation [43] pins the DSIC payment rule down (up to a constant) from monotonicity of the allocation in the reported type; the multi-parameter extension replaces monotonicity by cyclic monotonicity [44], and Roberts' theorem [45] restricts unrestricted quasi-linear DSIC mechanisms to affine maximisers. Without a payment channel, the utility that DSIC would give agent i is pinned down directly by the allocation via the envelope integral $U_i(w_i) = U_i(\underline{w}) + \int_{\underline{w}}^{w_i} (-\overline{A}_i(s))\, ds$, where $\overline{A}_i(s) = \mathbb{E}_{\theta_{-i}}[A_i(s, \theta_{-i})]$ is the interim expected realised advantage. For the actual utility $V(\sigma) - w_i \overline{A}_i(w_i)$ to coincide with this envelope, the allocation would have to be very special—for instance a Roberts affine maximiser of a shared welfare measure. Failing that, the appropriate way to falsify DSIC of the no-transfer identity mechanism is to exhibit a concrete opponent-dependent shift in the best-response report.

Proposition 21 (Non-strategyproofness of sampling-based iDP under budget declaration)**.** *Consider the identity mechanism on the budget-declaration channel* Σ_i *of Theorem 1. Fix any player* i *with true type* $w_i > 0$*. There exist opponent declaration profiles* $\hat{\varepsilon}_{-i}^A, \hat{\varepsilon}_{-i}^B \in \prod_{j \neq i} \Sigma_j$ *such that*

$$\arg\max_{\hat{\varepsilon}_i \in \Sigma_i} u_i\big(\hat{\varepsilon}_i, \hat{\varepsilon}_{-i}^A; w_i\big) \;\neq\; \arg\max_{\hat{\varepsilon}_i \in \Sigma_i} u_i\big(\hat{\varepsilon}_i, \hat{\varepsilon}_{-i}^B; w_i\big).$$

In particular, no report $\hat{\varepsilon}_i^{\mathrm{DS}} \in \Sigma_i$ is a dominant strategy for player i, so the identity mechanism is not DSIC on the budget channel.

Proof. Fix i and $w_i > 0$. Choose two opponent profiles on opposite sides of the externality: $\hat{\varepsilon}_{-i}^{A}$ has opponents declaring high budgets (small externality) and $\hat{\varepsilon}_{-i}^{B}$ has opponents declaring low budgets (large externality). Denote the resulting mechanism-solution maps $\phi^K(\hat{\varepsilon}_i) := (\sigma^K, \{p_j^K\}, \{A_j^K\})$ under $\hat{\varepsilon}_{-i}^{K}$ for $K \in \{A, B\}$, and write $u_i^K(\varepsilon) := V(\sigma^K(\varepsilon)) - w_i A_i^K(\varepsilon)$. By Theorem 7, $\partial\sigma/\partial\hat{\varepsilon}_j < 0$ and $\partial p_i/\partial\hat{\varepsilon}_j < 0$ for every $j \neq i$; empirically (Fig. 5), $\partial A_i/\partial\hat{\varepsilon}_j \leq 0$. Hence pointwise on Σ_i, $\sigma^B(\varepsilon) > \sigma^A(\varepsilon)$ and $A_i^B(\varepsilon) \geq A_i^A(\varepsilon)$, and the two marginal-utility gradients $\partial_{\hat{\varepsilon}_i} u_i^K = V'(\sigma^K)\partial_{\hat{\varepsilon}_i}\sigma^K - w_i \partial_{\hat{\varepsilon}_i} A_i^K$ differ across K. The two best-response maps $\hat{\varepsilon}_i^{\star,K} := \arg\max u_i^K$ are therefore not point-wise equal.

The numerical witness of Section E confirms this at the $(|G|, E_b, I, \delta)$ scenario of Fig. 5 and $w_1 = 1.0$: against $\hat{\varepsilon}_2 = 4.0$ the best response is $\hat{\varepsilon}_1^{\star,A} \approx 2.919$ ($u_1 = +0.528$), while against $\hat{\varepsilon}_2 = 0.5$ it is $\hat{\varepsilon}_1^{\star,B} \approx 1.208$ ($u_1 = +0.212$). No single $\hat{\varepsilon}_1^{\mathrm{DS}}$ maximises u_1 against both opponent profiles. □

Numerical witness. On $|G_1| = |G_2| = 25\,000$, $E_b = 128$, $I = 1953$, $\delta = 10^{-12}$, and $w_1 = 1.0$, we sweep player 1's best response over the log-spaced budget grid $\Sigma = \{0.50, 0.78, 1.21, 1.88, 2.92, 4.54, 7.05, 10.96, 17.04, 26.49, 41.17, 64.0\}$ as $\hat{\varepsilon}_2$ ranges over the same grid. The best-response argmax takes six distinct values $\hat{\varepsilon}_1^{\star}(\hat{\varepsilon}_2) \in \{1.208, 1.878, 2.919, 4.537, 7.053, 10.963\}$ as $\hat{\varepsilon}_2$ varies, generally rising with $\hat{\varepsilon}_2$ (smaller externality lets player 1 afford a larger own budget) with feasibility-driven kinks at intermediate $\hat{\varepsilon}_2$. Every candidate dominant strategy is therefore beaten by at least one opponent profile; the witness compares $\hat{\varepsilon}_2 \in \{0.5, 4.0\}$, both of which lie inside the feasible region under the participation constraint (Theorem 4).

Why the classical escape hatches fail. The large-market recovery of [46] requires the influence of a single agent on the mechanism's outcome to vanish as $n \to \infty$. In the federated-learning silo setting of interest, the meaningful scale is small n (a handful of silos) with large $|G_i|$ (millions of records per silo). Each silo's declaration affects (C2) at $\Theta(1)$ magnitude, so the $O(1/n)$-influence hypothesis is not met. The impossibility of truthful mechanisms for unbounded privacy valuations [47], [48] is not the direct obstacle here—utility is bounded (Section B) and weights are observable—but the coupled-σ structure that generates the externality is: the positive results of [49] model the privacy cost as vanishing when the agent's report has no influence on the outcome, whereas in the coupled PEG the report always shifts σ.

Setup for a type-channel Myerson analysis. A repair of the mechanism would elicit the type w_i rather than the budget $\hat{\varepsilon}_i$. Let $g\colon \prod_i \mathbb{R}_{>0} \to \mathcal{O}$ be the allocation. Truthful reporting in the Bayesian sense (BIC) and individual rationality (IR) then read

$$\overline{U}_i(w_i) - \overline{T}_i(w_i) \;\geq\; \overline{U}_i(\hat{w}_i; w_i) - \overline{T}_i(\hat{w}_i) \qquad \text{(BIC)}$$
$$\overline{U}_i(w_i) - \overline{T}_i(w_i) \;\geq\; \underline{u}_i(w_i) \qquad \text{(IR)}$$

for all $w_i, \hat{w}_i$, where $\overline{U}_i(\hat{w}_i; w_i) = \mathbb{E}_{\theta_{-i}}[V(\sigma(g(\hat{w}_i, \theta_{-i}))) - w_i A_i(g(\hat{w}_i, \theta_{-i}))]$ and $\overline{T}_i(\hat{w}_i) = \mathbb{E}_{\theta_{-i}}[t_i(\hat{w}_i, \theta_{-i})]$. With a payment channel, the envelope theorem then reduces BIC to monotonicity of $\overline{A}_i(w_i)$ in w_i. Three structural features distinguish this problem from a classical Myerson auction and keep the full theorem open: the mechanism must choose a common σ shared across all players; the interim advantage $\overline{A}_i$ depends on all θ_{-i} through the mechanism's coupling; and PEG has no monetary transfer, so the natural in-kind instrument is excess realised vulnerability, itself bilaterally constrained by the iDP contract.

Scope. Theorem 21 is a negative result on the budget channel; the constructive optimal-mechanism analysis on the type channel is left to future work.

Appendix F
Repeated Bayesian extension (BR-PEG)

The main paper analyses a one-shot full-information game. We have so far left aside two natural extensions: (i) repetition across rounds, and (ii) private knowledge of the players' utility weights. Both extensions have a clean home in the literature on repeated games with incomplete information, and this appendix sketches how the PEG specialises to a standard template in that line of work. We define the resulting game formally (the *Bayesian Repeated PEG*, BR-PEG) and report a small simulation that shows the qualitative behaviour the existing theory predicts.

Position in the literature. The classical framework for a repeated zero-sum game with one-sided private information is [37]; their central question is how the informed player should manage the information it reveals through its actions. A closely related strand, motivated by security applications, considers a leader who commits to a strategy each round and a follower of unknown type who best-responds; the leader observes the follower's response and refines its belief over the follower's payoff function [38]–[40]. Theoretical guarantees from [41] show that, under a mild absolute-continuity condition on the prior, rational Bayesian updating in such games leads to play that converges to a Nash equilibrium of the repeated game. A separate but related body of work studies privacy as a transacted good with heterogeneous private valuations [48], and sender-receiver problems in which a privileged party commits to an information structure [50]; both are conceptually close but solve different mechanism-design problems.

The BR-PEG is a direct instance of the Stackelberg-learning template of [38]–[40] applied to the iDP mechanism of Theorem 1. The leader is the (malicious) player M who commits to a budget ε_M and a sample count n_M each round; the follower is the (myopic) player H who best-responds with ε_H given her private type $\theta_H = (w_H, n_H)$; M observes (ε_H, u_H) and

Bayes-updates a posterior over θ_H. The differences from the standard template are minor: the follower's action space is the continuous-ish interval $[\varepsilon^{\min}, \varepsilon^{\max}]$ rather than a discrete set of attack targets, and the per-round payoff is delivered by the iDP mechanism rather than by a tabular game.

The game. Two players, H (honest) and M (malicious). H has a private type $\theta_H = (w_H, n_H)$ drawn from a finite known set Θ at $t = 0$; M does not know θ_H but knows Θ. Per round $t \in \{1, \dots, T\}$:

- M observes the history $\mathcal{H}_{t-1}$ and picks (ε_M^t, n_M^t).
- H observes (ε_M^t, n_M^t) and best-responds with the $\varepsilon_H^t \in \mathcal{A}$ that maximises its one-shot utility $u_H = V(\sigma) - w_H A_H$ given θ_H.
- The mechanism returns $(\sigma^t, p_H^t, p_M^t, A_H^t, A_M^t)$.
- The pair (ε_H^t, u_H^t) becomes public; M Bayes-updates its posterior over θ_H.
- H accrues u_H^t; M accrues A_H^t (M is malicious—it wants H's vulnerability up).

M's total score is $\sum_t A_H^t$. The strategic content is the identification problem: by observing how H's ε_H^t responds to M's own choices, M can narrow down θ_H and then exploit that knowledge in the remaining rounds. The framework is a one-sided incomplete-information repeated game in the sense of [37] but with the informed player myopic, which reduces it to the Stackelberg-learning instance of [39].

Simulation. We run a small instance to confirm the qualitative behaviour the existing theory predicts. H's true type is $w_H^\star = 2$ (privacy-preferring); M's prior is uniform over $\Theta = \{w_H \in \{0.5, 1, 2\}\}$ (utility-preferring, balanced, privacy-preferring). All other parameters are the canonical ones of Section VI, with the two players' group sizes fixed at $n_H = n_M = 5000$. We compare three M strategies: a benign constant $\varepsilon_M = 64$ (a baseline that should produce negligible A_H), a naive constant $\varepsilon_M = 1$ (the fixed-action attack of Section VI-D), and a posterior-greedy Bayesian policy that picks the ε_M on the grid maximising the expected next- round A_H under M's current posterior.

Figure 12 shows the result. Under the Bayesian-greedy policy, M's posterior collapses onto H's true type within one round; from $t = 1$ onwards M plays $\varepsilon_M^\star \approx 3.08$, eliciting $\varepsilon_H \approx 19$ from H and delivering a per-round $A_H \approx 0.454$ —roughly twice the naive attack ($A_H \approx 0.22$ at $\varepsilon_M = 1$) and around fifty times the benign baseline ($A_H \approx 0.01$ at $\varepsilon_M = 64$). The optimal adversarial budget is therefore *not* the minimum—the numerical realisation of Theorem 18 with $\eta \approx 2.08$ against the naive-attack anchor $\varepsilon^{\min} = 1$: M's gain comes not from forcing σ down but from drawing out a more exposing best response from H, a manoeuvre that requires knowing H's type and is opened up precisely by the Bayesian updating. This matches the intuition from the Stackelberg-learning literature [39], [40], in which a leader equipped with type knowledge does strictly better than one committed to a fixed strategy.

Full proof of Theorem 18. Write $\bar{G}(\varepsilon_M; \pi) := \mathbb{E}_{\theta\sim\pi} G(\varepsilon_M; \theta)$ for the posterior-averaged score. The map $\varepsilon_M \mapsto \bar{G}(\varepsilon_M; \pi)$ is a finite mixture of the per-type scores, hence continuous in π under the total-variation topology and defined on the finite grid $\mathcal{A}$.

Step 1 (posterior collapses on the truth). By identifiability (I) and [41], $\pi^t \to \delta_{\theta_H^\star}$ π^0-a.s.; equivalently, $\|\pi^t - \delta_{\theta_H^\star}\|_{\mathrm{TV}} = 1 - \pi^t(\theta_H^\star) \to 0$. The Kalai–Lehrer merging conclusion specialises here to a finite type space with a full-support prior π^0 and a type-separating log-Gaussian likelihood on (ε_H^t, u_H^t): any $\theta \neq \theta_H^\star$ that produces a distinct best response at M's chosen ε_M^t contributes a strictly positive Kullback–Leibler divergence to the log-likelihood, and this divergence accumulates over rounds.

Step 2 (score converges uniformly to the truth-score). Because $G(\varepsilon_M; \theta)$ is uniformly bounded ($0 \le A_H \le 1$), for every $\varepsilon_M \in \mathcal{A}$,

$$\left|\bar{G}(\varepsilon_M; \pi^t) - G(\varepsilon_M; \theta_H^\star)\right| \;\le\; \sum_{\theta \neq \theta_H^\star} \pi^t(\theta) \;=\; 1 - \pi^t(\theta_H^\star) \;\longrightarrow\; 0.$$

Since $\mathcal{A}$ is finite, the convergence is uniform in ε_M.

Step 3 (argmax converges to the truth-argmax). Let $\varepsilon_M^\star := \varepsilon_M^\star(\theta_H^\star)$ and $\Delta := G(\varepsilon_M^\star; \theta_H^\star) - \max_{\varepsilon_M \in \mathcal{A}\setminus\{\varepsilon_M^\star\}} G(\varepsilon_M; \theta_H^\star)$. By (A), $\Delta > 0$. Choose t_0 such that $1 - \pi^t(\theta_H^\star) < \Delta/2$ for all $t \ge t_0$; such t_0 exists by Step 1. For $t \ge t_0$ and any $\varepsilon_M \neq \varepsilon_M^\star$,

$$\begin{aligned}\bar{G}(\varepsilon_M^\star; \pi^t) - \bar{G}(\varepsilon_M; \pi^t) \;&\ge\; G(\varepsilon_M^\star; \theta_H^\star) - G(\varepsilon_M; \theta_H^\star) - 2(1 - \pi^t(\theta_H^\star)) \\ &\ge\; \Delta - \Delta \;>\; 0\end{aligned}$$

whenever $G(\varepsilon_M; \theta_H^\star) \le G(\varepsilon_M^\star; \theta_H^\star) - \Delta$, which holds by the definition of Δ. Hence $\varepsilon_M^{\mathrm{greedy}}(t) = \varepsilon_M^\star$ for all $t \ge t_0$.

Step 4 (separation from $\varepsilon^{\min}$). By (A), $\eta := \varepsilon_M^\star - \varepsilon^{\min} > 0$; substituting into the conclusion of Step 3 gives $\varepsilon_M^{\mathrm{greedy}}(t) \ge \varepsilon^{\min} + \eta$ for every $t \ge t_0$.

Assumption (A) is checked numerically for the demo instance above: the argmax of $G(\cdot; \theta_H^\star)$ on the mechanism grid lies at $\varepsilon_M^\star \approx 3.08$, well above the grid floor $\varepsilon^{\min} = 0.5$ and the naive-attack anchor $\varepsilon^{\min} = 1$. A closed-form monotone characterisation of $\varepsilon_H^{\mathrm{BR}}(\cdot; \theta)$ is not available—the honest player's grid-argmax reverses sign across the interval as the sampling floor releases—so we treat (A) as a numerically-verified assumption tied to the concrete $(V, \Theta, \text{expected batch}, \delta)$ of Section VI.

What this leaves open. The simulation is illustrative rather than exhaustive. Two technical points sit just outside the scope of this paper but are natural follow- ups. The first is the case where H is also strategic across rounds (a fully two-sided repeated game in the sense of [37]), in which H would manage what it reveals through its action sequence. The second is regret-bound analysis along the lines of [38]: how many rounds does M need before its per-round A_H is within ε of the full-information optimum? Both extensions are well-studied in the Stackelberg-learning literature; we have not pursued them here in order to keep the focus on the one-shot phenomenology, but the framework above slots into either with no further modification.

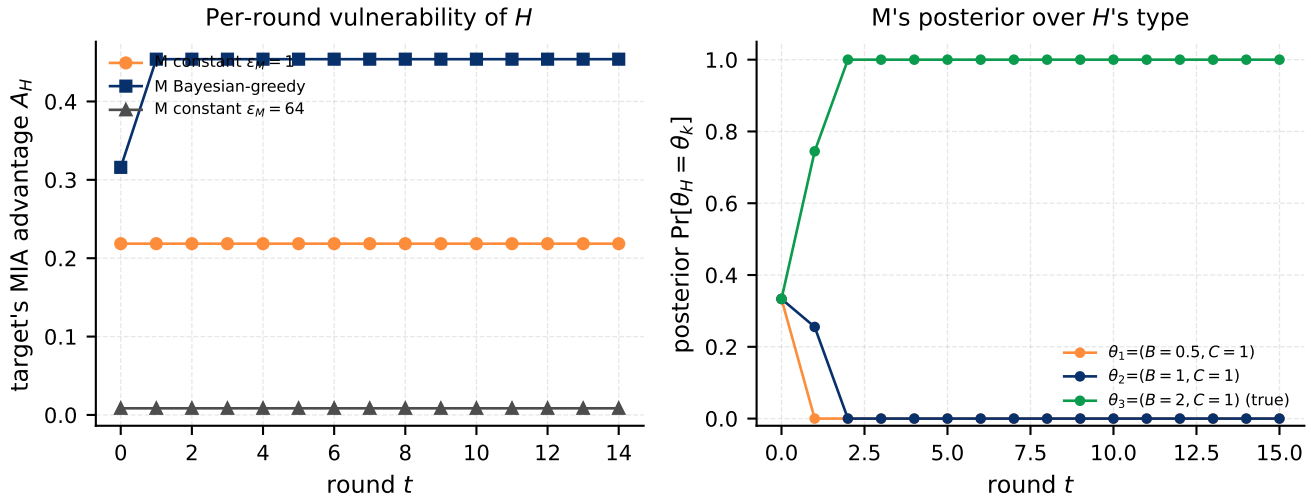


Fig. 12: **BR-PEG demo.** Left: per-round realised MIA advantage A_H of the honest player under three M policies. Right: M's posterior over H's type under the Bayesian-greedy policy; the posterior collapses to the true type within one round.